MARKET
RESEARCH AND
ANALYSIS

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NETFLIX OR ILLEGAL STREAMING PLATFORMS? WHAT DO PEOPLE CHOOSE AND WHY?

Analytical comparison between Netflix and Illegal streaming platforms

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INTRODUCTION

Whether someone is interested in tv series or movies, action or drama, documentaries or sci-fi, it is nowadays common practice to use streaming platforms as a form of access to this kind of entertainment. A streaming platform is an on-demand online entertainment source for TV shows, movies, and other streaming media, i.e. multimedia that is delivered and consumed in a continuous manner from a source, with little or no intermediate storage in network elements. This is just a fancy way of saying that streaming platforms deliver directly from the platform to the consumer.

The question is: *Why are we interested in them and what exactly are we looking for by studying them?*

We are interested in them because everyone knows them and uses them due to the growing popularity of legal and accessible streaming platforms such as Netflix, Amazon Prime Video, or Disney+, which makes them an obviously fascinating topic to elaborate on. Additionally, legal platforms are different from illegal ones and that creates a very nice ground for a detailed comparison between the two categories. This perfectly explains why we will deepen on the matter and study the behavior people have towards them.
In this analysis, we will focus on what people find convenient about one type of platform or the other, we will proceed to try to understand if there are connections between these characteristics and the choice of the platforms made by individuals, and finally, we will draw our conclusions on why some platforms are preferable compared to others and what is considered dislikable about them.

1.1 SURVEY DESCRIPTION

We created and designed the survey using Google Forms tool. We have been able to collect **111** responses from our families, friends and University colleagues from all over the world. One of our goals in this part was to receive as many responses from different socio-demographic backgrounds as possible. We decided to conduct the survey in English. Eventually, we analyzed our collected data with one of the most widely used statistical analysis and data visualization software packages SAS.

The survey included 4 different types of questions:

- *Sociodemographic questions,* used to understand personal aspects and social backgrounds of the sample.

- *Quantitative questions,* in order to comprehend the opinions of the respondents and how they value specific attributes of our topic on a scale from 1 to 7; 1 representing *"I hate it"* and 7 representing *"I love it".*

- *Behavioural questions, which are* based on the habits of our participants of the survey related to the topic.

- *Lifestyle questions, in order to* perceive the attitudes and daily routines of our respondents.

1.2 SURVEY QUESTIONS

Sociodemographic questions:

Q1: Age

- < 18
- 18-25
- 26-35
- 36-50
- \+ 50

Q2: Gender

- Female
- Male
- Non-binary
- I prefer not to say

Q3: Status

- Student
- Worker
- Student and Worker
- Unemployed
- Retiree

Q4: Level of education

- Elementary School
- High School
- Bachelor's Degree
- Master's Degree
- Ph.D. or higher
- Trade school

Q5: Country of residence (a list with all countries was given to respondents to select from)

Q6: Country of origin (a list with all countries was given to respondents to select from)

Q7: Household monthly income

- < 1000
- 1000-2000
- 2001-3000
- 3001-4000
- > 4001

Q8: Family members

- 1
- 2
- 3
- 4
- + 4

Q9: Minors in the family

- Yes
- No

<u>Qualitative or importance questions (scale 1-7):</u>

Q1: Rate the following Netflix features from (1) I hate it to (7) I love it, (4) is I don't care

- **Q1.1** Wide content
- **Q1.2** Convenience (no ads, high resolution)
- **Q1.3** Generated recommendations
- **Q1.4** Customer service
- **Q1.5** Quality of Netflix productions
- **Q1.6** Different subscription plans
- **Q1.7** Access to many languages in both audio and subtitles
- **Q1.8** Possibility of watching offline
- **Q1.9** Cost of the subscription
- **Q1.10** Longer waiting time due to copyright
- **Q1.11** Parental control

Q2: Rate the illegal streaming features from (1) I hate it to (7) I love it, (4) is I don't care

- **Q2.1** Wider content choices
- **Q2.2** Saving money
- **Q2.3** Access to other countries' productions
- **Q2.4** Search for the preferred language
- **Q2.5** Pop-up and explicit pop-up
- **Q2.6** Catching malware and viruses
- **Q2.7** Possibility of downloading the episodes
- **Q2.8** Watch simultaneously with friends without multiple accounts
- **Q2.9** Less waiting time
- **Q2.10** No content control

<u>Behavioral questions</u>:

Q1: What streaming services do you use?

- Netflix
- Amazon Prime Video
- Disney +
- HBO Max
- Hulu
- NowTV
- Illegal Streaming Platforms

Q2: How many hours a week you spend using legal streaming services (Netflix, Disney +, Amazon Prime, Hulu, HBO...)?

- 30 mins
- 1-4 hours
- 5+ hours
- I don't use streaming services

Q3: How many hours a week you spend watching Netflix??

- 30 mins
- 1-4 hours
- 5+ hours
- I don't watch Netflix

Q4: How many hours a week you spend using illegal streaming services?

- 30 mins
- 1-4 hours
- 5+ hours
- I don't use illegal streaming services

Q5: If you do use illegal streaming services, why do you do so?

- Better choices compared to Netflix, Disney+ or other legal streaming services
- Prices for legal streaming services are too high
- In my country, legal streaming services are not available
- I randomly watch series and movies so I don't see the need to have a monthly subscription

Q6: Do you share your Netflix account with anybody?

- Family
- Friends
- Nobody
- I don't have an account

Q7: What do you mainly watch?

- Tv series
- Movies

Q8: What kind of series/movies do you watch? (a list with all the mainly genres was given in order to respondents to select from)

Q9: Are you sensitive to copyright rights/issues?

- Yes
- No

<u>Lifestyle questions</u>:

Q1: Are you sensitive to price?

- Yes
- No

Q2: Do you like to stay indoor?

- → Yes
- → No

Q3: Do you prefer a healthy lifestyle?

- → Yes
- → No

Q4: Do you consider leisure time important?

- → Yes
- → No

Q5: Do you consider yourself a risk-taking person?

- → Yes
- → No

Q6: Do you feel stressed in your daily life?

- → Yes
- → No

1.3 PREPARING DATA FOR SAS ANALYSIS

We created an Excel table with all our data and we changed the name of the variables to make the commands in SAS easier. To identify the sociodemographic variables we chose the letter 'd', 'q' for qualitative questions, for the behavioral variables the letter 'b', and 'l' for lifestyle ones. Then we saved the file as a text file and imported it into SAS to start the analysis.

SOCIO-DEMOGRAPHIC ANALYSIS

2.1 FREQ PROCEDURE

First of all, we have completed the frequency procedure with sociodemographic variables in order to understand our sample data and the distribution of variables. The command we used allowed us to calculate the number, percentages, and cumulative percentages of our respondents falling in a specific category of our categorical variables.

```
proc freq data=streaming;
tables d_1--d_9;
run;
```

From SAS results represented as tables, we can see that majority (66,67%) of our replies came from people that are aged between 18 to 25 years old, while percentages of replies from the other age segments (<18; 26-35; 36-50; 50+) are 4,5%, 18,92%, 5,41%, 4,5% respectively. In addition, most (59,46%) of our respondents identify themselves as females, 33,33% as males, 4,5% as non-binary, 1,8% as gender-fluid, and the rest of respondents preferred not to say their gender.

Considering our main topic we were expecting a higher percentage of replies from students, we have included the third question (d_3) about the status in our survey in order to verify and know the exact percentage of students in our sample. They made up for 43,24% in total, followed by workers (36,04%), and people who are both: students and workers (16,22%). Also, we collected replies from 3,6% of unemployed respondents and 0,9% retired.

In addition, we wanted to get information about the highest level of education that our respondents have completed. From the results, we can see that the majority of our respondents have completed either Bachelor's degree (38,74%) or High School (44,14%). The 13,51% of our data were collected from people who completed their Master's degrees. These results also confirm the fact that most of our respondents were students. There were also replies from respondents who completed only elementary school or trade school. 1,8%; 0,9% and 0,9% respectively. There were some respondents who completed Ph.D. or higher (0,9%).

Mainly we got responses from respondents living and coming from Italy (living: 67,57%, coming: 61,26%) and Lithuania (living:16,22% and coming: 15,32%). Nevertheless, as we mentioned before we tried to collect data from respondents all over the world. Therefore we received answers from other European Union countries such as Germany (residents: 0,9% and respondents coming: 2,7%), Belgium residents (2,7%), Dutch people (0,9%), and also from non-EU countries such as Australian residents (0,9%), United States (residents:1,8% and coming from 0,9%) and Japanese respondents (2,7%).

We also analyzed the monthly household income in Euros of our respondents. Results show that for 20,72% of respondents monthly household income is less than 1000 euro, for 33,33% between 1000 and 2000 euro, for 16,22% between 2001 and 3000 euro, for 17,12% between 3001 and 4000 euro, and for 12,61% more than 4000 euro. As expected the results are related to the respondents' occupations.

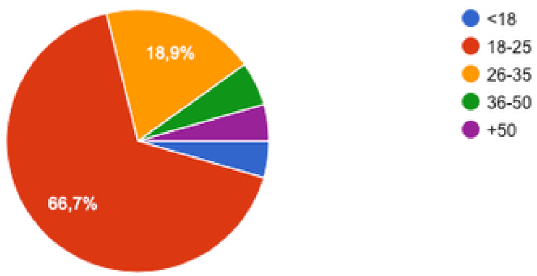
Following the reason that most of the respondents could be university students, we wanted to comprehend how their families are constituted. The largest group was families of four, representing 41,44%. The second most relevant were families of 3 (20,72%). Since the SAS did not recognize the +4 values, the following command was used to avoid *Frequency missing.*

```
proc freq data=streaming;
tables d_8/missing;
run;
```

Having information about the number of family members of our respondents we were also keen to know whether there are families that have siblings or kids younger than 18 years. We were considering the hypothesis that having younger members of the family could affect the choice between Netflix and illegal streaming platforms. Results showed that the majority of the respondents (72,07%) do not have siblings or children that are underaged in their families.

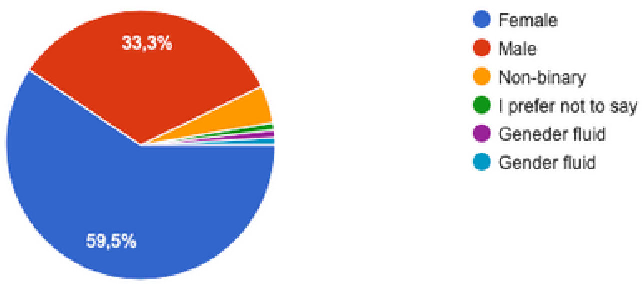
age

d_1				
d_1	Frequency	Percent	Cumulative Frequency	Cumulative Percent
+50	5	4.50	5	4.50
18-25	74	66.67	79	71.17
26-35	21	18.92	100	90.09
36-50	6	5.41	106	95.50
<18	5	4.50	111	100.00



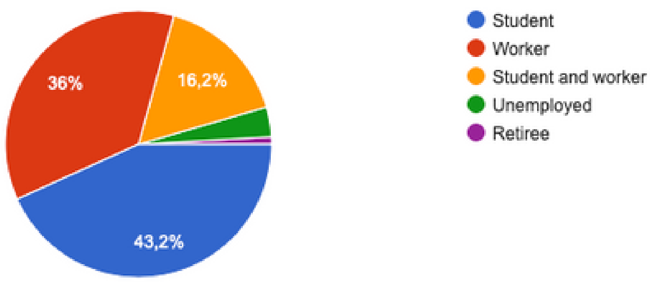
gender

d_2				
d_2	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	66	59.46	66	59.46
Gender fluid	2	1.80	68	61.26
I prefer not to say	1	0.90	69	62.16
Male	37	33.33	106	95.50
Non-binary	5	4.50	111	100.00



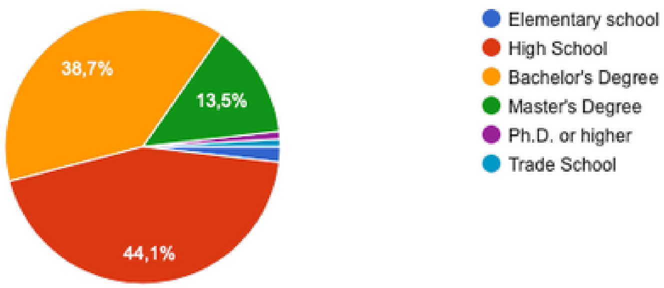
status

d_3				
d_3	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Retiree	1	0.90	1	0.90
Student	48	43.24	49	44.14
Student and worker	18	16.22	67	60.36
Unemployed	4	3.60	71	63.96
Worker	40	36.04	111	100.00



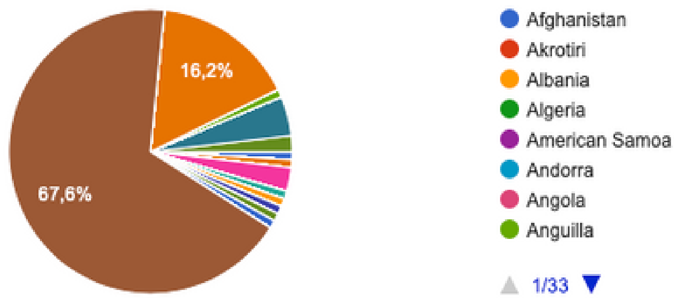
education

d_4				
d_4	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Bachelor's Degree	43	38.74	43	38.74
Elementary school	2	1.80	45	40.54
High School	49	44.14	94	84.68
Master's Degree	15	13.51	109	98.20
Ph.D. or higher	1	0.90	110	99.10
Trade School	1	0.90	111	100.00



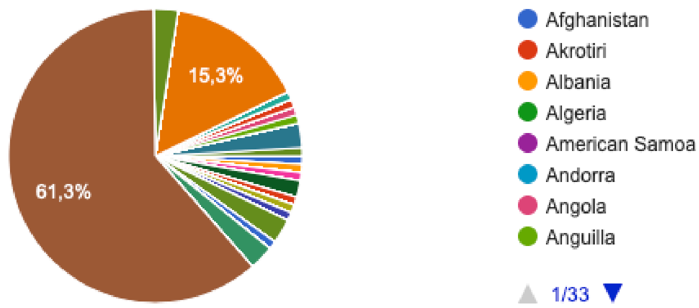
place of living

d_5				
d_5	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Afghanistan	1	0.90	1	0.90
Australia	1	0.90	2	1.80
Belgium	3	2.70	5	4.50
Canada	1	0.90	6	5.41
Denmark	1	0.90	7	6.31
France	1	0.90	8	7.21
Germany	1	0.90	9	8.11
Greece	1	0.90	10	9.01
Italy	75	67.57	85	76.58
Lithuania	18	16.22	103	92.79
Russia	1	0.90	104	93.69
United Kingdom	5	4.50	109	98.20
United States	2	1.80	111	100.00



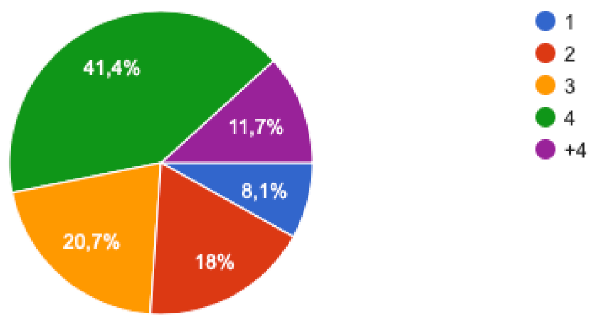
place of birth

d_6				
d_6	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Afghanistan	1	0.90	1	0.90
Albania	1	0.90	2	1.80
Belgium	1	0.90	3	2.70
Bosnia and Herzegovina	2	1.80	5	4.50
Brazil	1	0.90	6	5.41
Estonia	1	0.90	7	6.31
France	1	0.90	8	7.21
Germany	3	2.70	11	9.91
Greece	1	0.90	12	10.81
India	3	2.70	15	13.51
Italy	68	61.26	83	74.77
Japan	3	2.70	86	77.48
Lithuania	17	15.32	103	92.79
Netherlands	1	0.90	104	93.69
Poland	1	0.90	105	94.59
Romania	1	0.90	106	95.50
Russia	1	0.90	107	96.40
United Kingdom	3	2.70	110	99.10
United States	1	0.90	111	100.00



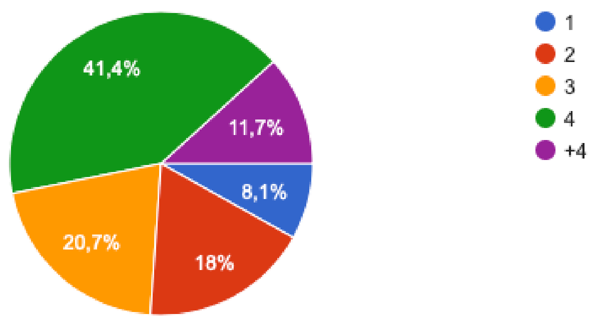
income

d_7				
d_7	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1000 -2000	37	33.33	37	33.33
2001-3000	18	16.22	55	49.55
3001-4000	19	17.12	74	66.67
< 1000	23	20.72	97	87.39
> 4001	14	12.61	111	100.00



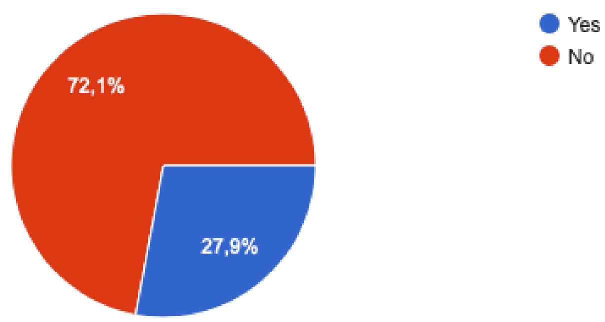
family members

d_8				
d_8	Frequency	Percent	Cumulative Frequency	Cumulative Percent
.	13	11.71	13	11.71
1	9	8.11	22	19.82
2	20	18.02	42	37.84
3	23	20.72	65	58.56
4	46	41.44	111	100.00



minors in family

d_9				
d_9	Frequency	Percent	Cumulative Frequency	Cumulative Percent
No	80	72.07	80	72.07
Yes	31	27.93	111	100.00



We can conclude that our sample is a **convenience sample**, that does not represent the total population because we mainly collected responses from young people.

IMPORTANCE QUESTIONS AND PRINCIPAL COMPONENT ANALYSIS

3.1 MEANS PROCEDURE

We began our analysis of the Importance data with the **means procedure**, which allows us to see main statistical measures such as **mean, standard deviation, minimum and maximum.** It calculates the average value for each variable, in this way we can understand which variables are most important in our sample.
In addition, **standard deviation** measures the dispersion of the values around the mean: the further the values are, the greater the standard deviation.

```
proc means data=streaming;
var q:;
run;
```

The SAS System

The MEANS Procedure

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
q_1	q_1	111	5.0630631	1.6252512	1.0000000	7.0000000
q_2	q_2	110	5.5090909	1.7333504	1.0000000	7.0000000
q_3	q_3	111	4.7567568	1.8498954	1.0000000	7.0000000
q_4	q_4	111	4.3513514	1.6217199	1.0000000	7.0000000
q_5	q_5	111	4.8648649	1.8610187	1.0000000	7.0000000
q_6	q_6	111	4.6486486	1.6604964	1.0000000	7.0000000
q_7	q_7	110	5.4636364	1.8408079	1.0000000	7.0000000
q_8	q_8	111	5.4414414	1.6716061	1.0000000	7.0000000
q_9	q_9	111	3.7477477	1.9608369	1.0000000	7.0000000
q_10	q_10	111	3.4684685	1.5244660	1.0000000	7.0000000
q_11	q_11	111	4.0900901	1.5228535	1.0000000	7.0000000
q_12	q_12	94	5.5319149	1.9438468	1.0000000	7.0000000
q_13	q_13	94	5.4787234	1.9605174	1.0000000	7.0000000
q_14	q_14	94	5.6382979	1.7588551	1.0000000	7.0000000
q_15	q_15	94	4.4787234	1.9659944	1.0000000	7.0000000
q_16	q_16	93	2.1612903	1.7463647	1.0000000	7.0000000
q_17	q_17	93	1.9139785	1.5649414	1.0000000	7.0000000
q_18	q_18	94	4.2659574	1.9901669	1.0000000	7.0000000
q_19	q_19	92	4.4347826	2.0929203	1.0000000	7.0000000
q_20	q_20	93	4.5806452	2.0500402	1.0000000	7.0000000
q_21	q_21	93	3.7634409	1.9693182	1.0000000	7.0000000

From the table above we can see that Netflix's convenience of having **no advertisement** and **high resolution** (q_2) and the access to other countries' productions (q_14) of illegal platforms are the most important variables to describe the choice between Netflix and illegal platforms in our sample. Because their means are respectively 5.5 and 5.53 out of 7. Our respondents are not that concerned about catching malware or viruses (q_17) while using illegal streaming platforms.
Variables q_10 **(longer waiting time due to copyright)** and q_11 (**parental contro**l) have the lowest standard deviation, which means that respondents' responses did not vary much around the mean.
The **biggest SD** (2.09) comes from the variable of watching simultaneously with friends without multiple accounts (q_19), which represents the fact that most people disagree about their judgment of this illegal platforms' feature. From the results, we can also see that people also disagree a lot about less waiting time for illegal streaming platforms (2.05).

3.2 PRINCIPAL COMPONENT ANALYSIS

Principal component analysis is a factor transformation of our original dataspace with which we aim to reduce the number of dimensions of a dataspace while preserving the essential part of variability. The outcomes of this analysis are: a **correlation matrix, an eigenvalue table, and an eigenvector table.**
From the input dataset consisting of "n" rows and "p" variables, "p principal components" can be computed, each one of which is a linear combination of the original variables, with coefficients equal to the eigenvectors of the correlation matrix.
We can start with the **correlation**, which is a measure that spans from -1 to +1 and describes the linear relationship between two variables.
Two variables move in the same direction (when one increases, the other goes up as well) if the correlation coefficient is positive, and in opposite direction (when one increases and the other decreases) the coefficient is negative. If the correlation is +1 or -1 we can see the perfect linear dependency, while if it is equal to 0 the two variables are not correlated.
Correlation then is the ratio of the covariance (measurement of confusion on a data space) of the two variables to the product of their standard deviations.

We used following SAS procedure:

```
proc princomp data=streaming out=correlation_matrix;
 var q_1--q_21;
 run;
```

Correlation Matrix																						
		q_1	q_2	q_3	q_4	q_5	q_6	q_7	q_8	q_9	q_10	q_11	q_12	q_13	q_14	q_15	q_16	q_17	q_18	q_19	q_20	q_21
q_1	q_1	1.0000	0.6333	0.6314	0.5060	0.6152	0.5567	0.5023	0.6317	0.1776	0.2182	0.3480	0.2554	0.1790	0.2259	0.1828	-.2370	-.2806	0.1177	0.1191	0.1715	0.0280
q_2	q_2	0.6333	1.0000	0.6022	0.5364	0.4803	0.5697	0.4719	0.5481	0.4034	0.1257	0.2746	0.1741	0.1173	0.1913	0.1436	-.2029	-.2725	0.0187	0.0528	0.1785	-.0230
q_3	q_3	0.6314	0.6022	1.0000	0.6195	0.4668	0.5853	0.3870	0.5080	0.2194	0.1447	0.3275	0.2418	0.1124	0.3177	0.1633	-.1922	-.2059	0.1013	0.0836	0.2133	0.0174
q_4	q_4	0.5060	0.5364	0.6195	1.0000	0.5500	0.5280	0.3193	0.4829	0.3033	0.2853	0.4075	-.0187	-.0748	0.0837	0.1508	-.0276	-.0248	-.0512	-.0041	0.1628	0.0562
q_5	q_5	0.6152	0.4803	0.4668	0.5500	1.0000	0.5567	0.2750	0.4585	0.1360	0.2783	0.1712	0.0202	-.0104	-.0086	0.2361	-.0639	-.1319	0.0546	0.0021	0.1445	0.0121
q_6	q_6	0.5567	0.5697	0.5853	0.5280	0.5567	1.0000	0.4755	0.5476	0.3559	0.2040	0.3214	0.1501	0.0602	0.1940	0.1882	-.1677	-.2043	-.0049	0.0318	0.1094	-.0435
q_7	q_7	0.5023	0.4719	0.3870	0.3193	0.2750	0.4755	1.0000	0.5698	0.1024	0.1589	0.2792	0.2791	0.2352	0.2549	0.0373	-.4248	-.3774	0.0307	0.0492	0.1619	0.0265
q_8	q_8	0.6317	0.5481	0.5080	0.4829	0.4585	0.5476	0.5698	1.0000	0.2274	0.1359	0.4412	0.3115	0.2866	0.3569	0.1563	-.2767	-.3412	0.1217	0.2435	0.2703	0.0082
q_9	q_9	0.1776	0.4034	0.2194	0.3033	0.1360	0.3559	0.1024	0.2274	1.0000	0.2838	0.1910	-.0288	-.0350	0.0185	0.0653	-.0593	-.0991	-.2362	-.0926	-.1405	-.0205
q_10	q_10	0.2182	0.1257	0.1447	0.2853	0.2783	0.2040	0.1589	0.1359	0.2838	1.0000	0.2793	-.1183	-.0953	-.1226	0.0192	-.0214	0.1139	-.0362	-.0696	-.1063	0.1542
q_11	q_11	0.3480	0.2746	0.3275	0.4075	0.1712	0.3214	0.2792	0.4412	0.1910	0.2793	1.0000	0.0796	0.1230	0.1454	0.1889	0.0347	-.0508	0.1037	0.1866	0.1915	0.1237
q_12	q_12	0.2554	0.1741	0.2418	-.0187	0.0202	0.1501	0.2791	0.3115	-.0288	-.1183	0.0796	1.0000	0.8269	0.8397	0.4696	-.1062	-.2041	0.4187	0.4517	0.5462	0.3694
q_13	q_13	0.1790	0.1173	0.1124	-.0748	-.0104	0.0602	0.2352	0.2866	-.0350	-.0953	0.1230	0.8269	1.0000	0.7824	0.3902	-.1378	-.2241	0.3802	0.3712	0.4305	0.2590
q_14	q_14	0.2259	0.1913	0.3177	0.0837	-.0086	0.1940	0.2549	0.3569	0.0185	-.1226	0.1454	0.8397	0.7824	1.0000	0.4578	-.0870	-.1483	0.3821	0.3901	0.5532	0.3141
q_15	q_15	0.1828	0.1436	0.1633	0.1508	0.2361	0.1882	0.0373	0.1563	0.0653	0.0192	0.1889	0.4696	0.3902	0.4578	1.0000	0.1829	0.0452	0.5705	0.5522	0.5777	0.3835
q_16	q_16	-.2370	-.2029	-.1922	-.0276	-.0639	-.1677	-.4248	-.2767	-.0593	-.0214	0.0347	-.1062	-.1378	-.0870	0.1829	1.0000	0.6759	0.2352	0.1777	0.1572	0.3113
q_17	q_17	-.2806	-.2725	-.2059	-.0248	-.1319	-.2043	-.3774	-.3412	-.0991	0.1139	-.0508	-.2041	-.2241	-.1483	0.0452	0.6759	1.0000	0.2347	0.1986	0.0681	0.2781
q_18	q_18	0.1177	0.0187	0.1013	-.0512	0.0546	-.0049	0.0307	0.1217	-.2362	-.0362	0.1037	0.4187	0.3802	0.3821	0.5705	0.2352	0.2347	1.0000	0.7787	0.6627	0.4385
q_19	q_19	0.1191	0.0528	0.0836	-.0041	0.0021	0.0318	0.0492	0.2435	-.0926	-.0696	0.1866	0.4517	0.3712	0.3901	0.5522	0.1777	0.1986	0.7787	1.0000	0.6652	0.4544
q_20	q_20	0.1715	0.1785	0.2133	0.1628	0.1445	0.1094	0.1619	0.2703	-.1405	-.1063	0.1915	0.5462	0.4305	0.5532	0.5777	0.1572	0.0681	0.6627	0.6652	1.0000	0.5597
q_21	q_21	0.0280	-.0230	0.0174	0.0562	0.0121	-.0435	0.0265	0.0082	-.0205	0.1542	0.1237	0.3694	0.2590	0.3141	0.3835	0.3113	0.2781	0.4385	0.4544	0.5597	1.0000

We can see the **highest positive correlation coefficient** (0.84) between wider content choices (q_12) and access to other countries' productions (q_14) of illegal streaming platforms. The **highest negative correlation** (-0.42) is between access to many languages in both audio and subtitles on Netflix(q_7) and pop-up and explicit pop-ups in illegal streaming platforms (q_16). It makes sense because people who are willing to access other countries' productions automatically could have a wider content choice in illegal streaming.

However, most correlation coefficients are positive, which means that our matrix is inflated by a positive correlation trend. And, since our variables have some linear dependency between each other we are not operating in Euclidean space (in which all variables are independent and orthogonal). As a consequence, we cannot use either the Ward method or the other algorithms based on distance measurements of our units to create clusters.

Our database should have a 0 correlation to be useful for our clusterization because each variable needs to have a specific meaning in characterizing the clusters.

In the following table we have **eigenvalues**.

Eigenvalues of the Correlation Matrix				
	Eigenvalue	Difference	Proportion	Cumulative
1	6.00963009	1.86196296	0.2862	0.2862
2	4.14766713	1.83347270	0.1975	0.4837
3	2.31419443	1.15573227	0.1102	0.5939
4	1.15846216	0.16497306	0.0552	0.6490
5	0.99348910	0.15116911	0.0473	0.6964
6	0.84231999	0.06317870	0.0401	0.7365
7	0.77914129	0.10161648	0.0371	0.7736
8	0.67752482	0.08318360	0.0323	0.8058
9	0.59434122	0.05579756	0.0283	0.8341
10	0.53854366	0.04835670	0.0256	0.8598
11	0.49018696	0.06782758	0.0233	0.8831
12	0.42235938	0.03745776	0.0201	0.9032
13	0.38490162	0.07654398	0.0183	0.9216
14	0.30835764	0.04095340	0.0147	0.9362
15	0.26740424	0.03196373	0.0127	0.9490
16	0.23544052	0.01485849	0.0112	0.9602
17	0.22058203	0.01994254	0.0105	0.9707
18	0.20063949	0.02104039	0.0096	0.9802
19	0.17959910	0.05775111	0.0086	0.9888
20	0.12184799	0.00848083	0.0058	0.9946
21	0.11336716		0.0054	1.0000

Each eigenvalue is the length of each principal component and thus it describes the variability associated with each PC. We have 21 PC since we have 21 variables. We can find the portion of the variability that each PC represents by dividing its length by the sum of all their lengths, which is equal to 21.

To understand which PCs are most important to the general global variability we have to compare each eigenvalue to the benchmark (the expected value which is equal to 1). According to this benchmark strategy we have to select only those principal components with eigenvalues similar to or bigger than 1.

In this case, we can select the Principal Components 1-2-3-4 and also 5, since 0,99 is very close to 1, representing 69% of the **variability** (as we can see in the cumulative column) passing from 21 dimensions space to a space with 5 dimensions.

We can describe the other principal components as **noise** generated by the mind of the respondents that confounds the similarity structure of our data.
In the **eigenvector** table, we see the correlation between each original variable and principal component.

Eigenvectors																						
		Prin1	Prin2	Prin3	Prin4	Prin5	Prin6	Prin7	Prin8	Prin9	Prin10	Prin11	Prin12	Prin13	Prin14	Prin15	Prin16	Prin17	Prin18	Prin19	Prin20	Prin21
q_1	q_1	0.303402	-.166634	0.030287	-.178977	-.053124	0.111825	0.087096	-.074769	0.175677	-.043363	-.417555	0.282775	-.149299	0.423194	-.023631	-.226208	0.194126	-.444384	-.166835	0.091884	0.084385
q_2	q_2	0.277601	-.193997	0.034603	-.041854	0.228312	-.054924	-.113205	0.305231	0.085672	-.042857	-.310621	0.321558	0.385406	-.404458	-.146492	-.258896	-.263557	0.181334	0.030884	-.090832	0.002915
q_3	q_3	0.287238	-.153402	0.060074	-.123640	0.184700	-.097618	0.232059	0.044064	-.084697	-.582090	0.121959	0.166339	-.237847	0.046012	0.266259	0.224115	0.078537	0.320973	0.204533	0.235180	0.030672
q_4	q_4	0.235702	-.203670	0.262682	-.027993	0.106371	-.114176	0.211801	0.004234	-.339926	-.148823	0.116605	-.397806	0.352179	0.194399	-.184198	0.162423	-.308049	-.340474	-.113892	-.066294	-.063958
q_5	q_5	0.225084	-.180642	0.200315	-.296480	0.090986	0.428756	0.032126	-.281158	-.043616	0.291565	-.152932	-.171175	-.041100	-.118406	-.192408	0.272119	0.260714	0.266019	0.199145	-.258354	0.070452
q_6	q_6	0.269262	-.202926	0.084501	-.024179	0.165494	0.013560	-.042307	-.005175	0.201394	0.239638	0.564402	0.076511	-.458111	-.261101	-.176588	-.127415	-.182737	-.208310	-.120121	0.100712	-.000740
q_7	q_7	0.247239	-.139584	-.175597	-.019449	-.366303	-.002147	0.089026	0.393318	0.165007	0.309123	0.345664	0.191928	0.348321	0.242721	0.122677	0.262853	0.139457	0.038817	0.132566	-.034804	0.090235
q_8	q_8	0.312105	-.121491	-.057515	-.050161	-.161189	-.216696	-.050003	-.011852	0.193484	0.216737	-.204502	-.530894	-.104420	0.064754	0.365305	-.211286	-.144926	0.323013	-.166141	0.147994	-.174108
q_9	q_9	0.102446	-.183027	0.108829	0.583530	0.352541	-.031663	-.457859	0.204708	0.070616	0.001528	-.100155	-.112978	-.062711	0.135410	0.011442	0.259723	0.319181	-.047242	-.055990	-.019700	-.039459
q_10	q_10	0.075902	-.135717	0.273344	0.429214	-.443092	0.473983	0.039040	-.158990	0.137764	-.269375	0.048331	-.042115	0.109329	-.248007	0.234067	-.138229	-.079711	-.106606	0.082562	0.019910	0.033887
q_11	q_11	0.195096	-.059513	0.174844	0.240243	-.386883	-.586787	-.001953	-.372793	-.203883	0.081805	-.055770	0.260698	-.073772	-.097944	-.219469	0.031782	0.143798	0.095368	0.042511	-.133912	-.063832
q_12	q_12	0.247146	0.258756	-.269525	0.170976	0.127777	0.102100	0.196830	-.048538	0.083773	0.007301	-.027495	0.036984	-.056423	0.047738	-.014656	-.016021	-.058009	-.143544	0.363979	-.226613	-.692319
q_13	q_13	0.210211	0.242448	-.309151	0.236409	0.051695	0.064657	0.172569	-.239209	0.165683	0.067437	-.149283	-.039761	0.143785	-.113487	-.299326	0.313661	-.121338	0.027835	-.084827	0.553813	0.211927
q_14	q_14	0.252475	0.234944	-.237541	0.207661	0.196992	-.053119	0.268730	-.079614	0.038869	-.097372	0.127102	-.116760	0.019590	-.014668	0.114042	-.225898	0.147245	0.053258	-.200122	-.522519	0.467246
q_15	q_15	0.208564	0.248286	0.126074	0.002553	0.178179	0.177283	-.371597	-.343453	-.249268	0.111299	0.267723	0.227149	0.260039	0.350362	0.143814	-.261314	-.098563	0.212085	-.012629	0.165148	-.021666
q_16	q_16	-.083431	0.196553	0.440563	0.038575	0.259555	-.180902	0.239875	-.086578	0.183937	0.343944	-.137465	0.171115	0.044800	-.082313	0.490317	0.232330	-.168745	-.189937	0.116145	-.008644	0.109997
q_17	q_17	-.116351	0.173728	0.456307	0.043315	0.072576	-.072531	0.286898	0.108673	0.426562	-.069076	0.139729	-.105107	0.153708	0.177922	-.359581	-.260929	0.276030	0.250533	-.007811	0.122028	-.132913
q_18	q_18	0.160026	0.343949	0.133410	-.241992	-.154260	0.039574	-.237156	0.001003	0.238314	-.246020	0.032525	0.100077	0.030137	-.113863	0.027830	0.385818	-.027192	0.012439	-.554769	-.213896	-.227631
q_19	q_19	0.177432	0.327541	0.117801	-.134117	-.154485	-.132821	-.380809	0.108028	0.202629	-.164705	-.047306	-.204731	-.137848	0.106748	-.162075	0.008766	-.218240	-.133563	0.530285	-.081488	0.329827
q_20	q_20	0.230776	0.304440	0.075490	-.155663	-.026637	-.040809	-.033278	0.204432	-.316480	0.049983	0.028245	-.157623	0.102657	-.405908	0.133014	-.156780	0.521301	-.279890	0.040531	0.281352	-.045678
q_21	q_21	0.115949	0.276419	0.215661	0.209580	-.151463	0.230736	0.188249	0.450074	-.393068	0.177171	-.169072	0.095391	-.368122	0.130371	-.124216	0.008703	-.216315	0.208875	-.146066	-.001768	0.046682

Looking at the correlation between our variables and PC1 we see that we have almost no trade-off, in fact when one variable increases almost all the others do the same. We have **size effect**, as also the positive trend in the correlation matrix suggested to us.

3.3 SIZE EFFECT

Size effect is a very common phenomenon in these types of surveys about opinions. It is the general trend of our measurement and it creates distortion and inflation in the data set.
This phenomenon is a consequence of the subjectivity of the people about the perception they have of the scale when they are rating, which depends on the average level in the mind of people participating and is related to emotional and cultural variables.
We need to eliminate this distortion (and the size effect) by transforming our original variables based on a scale from 1 to 7 into **new variables**, with a new scale from -1 to +1, in which -1 is the minimum, 0 is the average and +1 is the maximum.

In SAS we used **array function** to create the new variables from the old ones and the **if statement** to generate a code according to which the variables were to be changed. The command was used:

```
data streaming2;
set streaming;
avgi=mean(of q_1-q_21);
mini=min (of q_1-q_21);
maxi=max(of q_1-q_21);
array a1 q_1-q_21;
array a2 new_q_1-new_q_21;
do over a2;
if a1>avgi then a2=(a1-avgi)/(maxi-avgi);
if a1<avgi then a2=(a1-avgi)/(avgi-mini);
if a1 =. then a2=0;
end;
label new_q_1='Wide content';
label new_q_2='Convenience';
label new_q_3='Recommendations';
label new_q_4='Customer service';
label new_q_5='Quality';
label new_q_6='Plans';
label new_q_7='Languages';
label new_q_8='Offline watching';
label new_q_9='Cost';
label new_q_10='Longer waiting';
label new_q_11='Parental control';
label new_q_12='Wider content';
label new_q_13='Saving money';
label new_q_14='Other countries';
label new_q_15='Search language';
label new_q_16='Pop-up';
label new_q_17='Malware/viruses';
label new_q_18='Downloading';
label new_q_19='Friends';
label new_q_20='Less waiting';
label new_q_21='No content control';
run;
```

After changing our previous variables we performed again the means procedure.

```
proc means data=streaming2;
var new_q:;
run;
```

The SAS System

The MEANS Procedure

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
new_q_1	Wide content	104	0.3155903	0.6002676	-1.0000000	1.0000000
new_q_2	Convenience	104	0.5099735	0.5880187	-1.0000000	1.0000000
new_q_3	Recommendations	104	0.1912457	0.6514416	-1.0000000	1.0000000
new_q_4	Customer service	103	0.0130558	0.5975479	-1.0000000	1.0000000
new_q_5	Quality	106	0.2571689	0.6439013	-1.0000000	1.0000000
new_q_6	Plans	105	0.1734896	0.5866702	-1.0000000	1.0000000
new_q_7	Languages	106	0.4810362	0.6336371	-1.0000000	1.0000000
new_q_8	Offline watching	105	0.4734892	0.6090724	-1.0000000	1.0000000
new_q_9	Cost	105	-0.1641991	0.6852187	-1.0000000	1.0000000
new_q_10	Longer waiting	105	-0.3044582	0.5023821	-1.0000000	1.0000000
new_q_11	Parental control	105	-0.0929551	0.5696000	-1.0000000	1.0000000
new_q_12	Wider content	106	0.4571634	0.6319119	-1.0000000	1.0000000
new_q_13	Saving money	105	0.4421394	0.6358192	-1.0000000	1.0000000
new_q_14	Other countries	105	0.5053048	0.5527276	-1.0000000	1.0000000
new_q_15	Search language	103	0.1283723	0.6340076	-1.0000000	1.0000000
new_q_16	Pop-up	106	-0.5702534	0.5959092	-1.0000000	1.0000000
new_q_17	Malware/viruses	107	-0.6350119	0.5453148	-1.0000000	1.0000000
new_q_18	Downloading	105	0.0309431	0.6365972	-1.0000000	1.0000000
new_q_19	Friends	106	0.0783357	0.6638089	-1.0000000	1.0000000
new_q_20	Less waiting	105	0.1434639	0.6543631	-1.0000000	1.0000000
new_q_21	No content control	105	-0.1282944	0.6302170	-1.0000000	1.0000000

Compared to the old variables, there is a correspondence between the most and least important variables. In fact, the rank in terms of relative importance does not change. That is, transforming the data space does not change the relative importance between variables, it just eliminates the size effect of it: the convenience of Netflix and access to other countries' production of illegal streaming platforms are still the most important variables with the least standard deviations while the danger of malware or viruses is the least important for our sample.

At this point, we proceeded with the **principal component analysis** (PCA) to select the essential part of variability to describe our sample. The factor transformation of our original data with principal component analysis allows us to observe a dataspace by transforming it in an orthogonal space, in which we can use the Euclidean distance. We can prove it by computing the correlation coefficients between our Principal Components.

```
proc princomp data =streaming2 out=coord;
var new_q_1-new_q_21;
run;
```

Simple Statistics

	new_q_1	new_q_2	new_q_3	new_q_4	new_q_5	new_q_6	new_q_7	new_q_8	new_q_9	new_q_10	new_q_11
Mean	0.3152424157	0.5069011154	0.1802897283	0.0171053410	0.2411101526	0.1851281622	0.5129722911	0.4645395261	-.1624925070	-.3232167391	-.0858851839
StD	0.5988730816	0.5915377414	0.6521031530	0.5990767339	0.6410351931	0.5717356481	0.6157113008	0.6133704619	0.6786989950	0.4856905929	0.5530928165

	new_q_12	new_q_13	new_q_14	new_q_15	new_q_16	new_q_17	new_q_18	new_q_19	new_q_20	new_q_21
Mean	0.4685554048	0.4355357033	0.5038268851	0.1296308161	-.5762763741	-.6399960911	0.0269512045	0.0814076392	0.1346115719	-.1189958768
StD	0.6233390387	0.6388519390	0.5562114516	0.6370091652	0.6006453925	0.5490278127	0.6365366043	0.6618450764	0.6580318334	0.6330723818

Correlation Matrix

		new_q_1	new_q_2	new_q_3	new_q_4	new_q_5	new_q_6	new_q_7	new_q_8	new_q_9	new_q_10	new_q_11	new_q_12	new_q_13	new_q_14	new_q_15	new_q_16	new_q_17	new_q_18	new_q_19	new_q_20	new_q_21
new_q_1	Wide content	1.0000	0.3520	0.4257	0.2864	0.3775	0.2723	0.2252	0.4077	0.0545	-.0379	-.0371	-.0501	-.0811	-.1082	-.0365	-.2072	-.2735	-.1368	-.1125	-.0904	-.1896
new_q_2	Convenience	0.3520	1.0000	0.3188	0.2091	0.2122	0.1870	0.2554	0.2762	0.1443	-.2056	-.1058	-.0961	-.1068	-.1439	-.1053	-.2500	-.2964	-.2435	-.1619	-.0411	-.3062
new_q_3	Recommendations	0.4257	0.3188	1.0000	0.4306	0.2631	0.4200	0.1363	0.2768	0.1252	0.0419	0.1139	-.0211	-.1212	-.0051	-.0867	-.1756	-.1604	-.1199	-.1050	-.0484	-.1665
new_q_4	Customer service	0.2864	0.2091	0.4306	1.0000	0.2697	0.3175	0.0479	0.1336	0.2403	0.1699	0.2524	-.2608	-.2752	-.1534	-.0167	-.0296	-.0005	-.2409	-.1785	-.0426	-.1094
new_q_5	Quality	0.3775	0.2122	0.2631	0.2697	1.0000	0.3277	0.0569	0.2161	-.0018	0.1476	-.1611	-.2180	-.2093	-.2896	0.0031	-.0837	-.1230	-.1477	-.1898	-.0849	-.1589
new_q_6	Plans	0.2723	0.1870	0.4200	0.3175	0.3277	1.0000	0.1340	0.2733	0.3010	0.2115	0.1194	-.1820	-.1992	-.0980	-.0689	-.1387	-.1359	-.2605	-.2563	-.1504	-.2345
new_q_7	Languages	0.2252	0.2554	0.1363	0.0479	0.0569	0.1340	1.0000	0.3961	0.0442	-.0376	0.1216	0.0332	0.1104	-.0499	-.1937	-.4895	-.4144	-.1153	-.0698	-.0508	-.1956
new_q_8	Offline watching	0.4077	0.2762	0.2768	0.1336	0.2161	0.2733	0.3961	1.0000	0.1175	-.0773	0.0751	0.0155	0.0930	-.0281	-.1156	-.2898	-.3368	-.0891	0.0422	0.0003	-.2278
new_q_9	Cost	0.0545	0.1443	0.1252	0.2403	-.0018	0.3010	0.0442	0.1175	1.0000	0.2094	0.0906	-.2108	-.1213	-.1443	-.1066	-.0599	-.0741	-.3743	-.2944	-.2868	-.1501
new_q_10	Longer waiting	-.0379	-.2056	0.0419	0.1699	0.1476	0.2115	-.0376	-.0773	0.2094	1.0000	0.2554	-.3386	-.2828	-.3397	-.1992	-.0844	0.0536	-.2131	-.2720	-.3180	-.0142
new_q_11	Parental control	-.0371	-.1058	0.1139	0.2524	-.1611	0.1194	0.1216	0.0751	0.0906	0.2554	1.0000	-.0161	-.0531	-.0008	0.0423	-.0761	-.1757	-.0554	0.0516	0.0385	-.0286
new_q_12	Wider content	-.0501	-.0961	-.0211	-.2608	-.2180	-.1820	0.0332	0.0155	-.2108	-.3386	-.0161	1.0000	0.7260	0.7562	0.3753	-.2257	-.3690	0.3140	0.3764	0.4035	0.1867
new_q_13	Saving money	-.0811	-.1068	-.1212	-.2752	-.2093	-.1992	0.1104	0.0930	-.1213	-.2828	-.0531	0.7260	1.0000	0.6622	0.2393	-.3279	-.3901	0.2360	0.2447	0.2658	0.0813
new_q_14	Other countries	-.1082	-.1439	-.0051	-.1534	-.2896	-.0980	-.0499	-.0281	-.1443	-.3397	-.0008	0.7562	0.6622	1.0000	0.3672	-.2047	-.3108	0.2236	0.2611	0.4494	0.1813
new_q_15	Search language	-.0365	-.1053	-.0867	-.0167	0.0031	-.0689	-.1937	-.1156	-.1066	-.1992	0.0423	0.3753	0.2393	0.3672	1.0000	0.1805	0.0002	0.4398	0.4568	0.5199	0.2964
new_q_16	Pop-up	-.2072	-.2500	-.1756	-.0296	-.0837	-.1387	-.4895	-.2898	-.0599	-.0844	-.0761	-.2257	-.3279	-.2047	0.1805	1.0000	0.6820	0.2099	0.0809	0.1469	0.3167
new_q_17	Malware/viruses	-.2735	-.2964	-.1604	-.0005	-.1230	-.1359	-.4144	-.3368	-.0741	0.0536	-.1757	-.3690	-.3901	-.3108	0.0002	0.6820	1.0000	0.2046	0.0886	0.0087	0.2235
new_q_18	Downloading	-.1368	-.2435	-.1199	-.2409	-.1477	-.2605	-.1153	-.0891	-.3743	-.2131	-.0554	0.3140	0.2360	0.2236	0.4398	0.2099	0.2046	1.0000	0.6673	0.5440	0.3238
new_q_19	Friends	-.1125	-.1619	-.1050	-.1785	-.1898	-.2563	-.0698	0.0422	-.2944	-.2720	0.0516	0.3764	0.2447	0.2611	0.4568	0.0809	0.0886	0.6673	1.0000	0.5884	0.3828
new_q_20	Less waiting	-.0904	-.0411	-.0484	-.0426	-.0849	-.1504	-.0508	0.0003	-.2868	-.3180	0.0385	0.4035	0.2658	0.4494	0.5199	0.1469	0.0087	0.5440	0.5884	1.0000	0.5057
new_q_21	No content control	-.1896	-.3062	-.1665	-.1094	-.1589	-.2345	-.1956	-.2278	-.1501	-.0142	-.0286	0.1867	0.0813	0.1813	0.2964	0.3167	0.2235	0.3238	0.3828	0.5057	1.0000

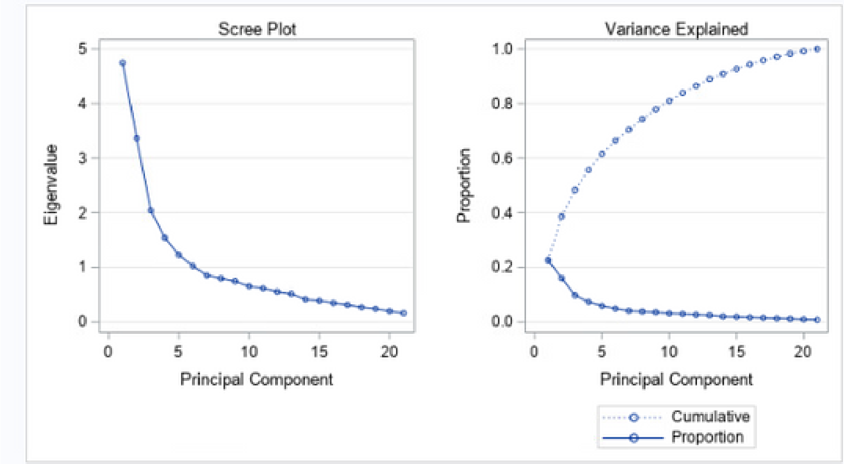
We observe that now, after the elimination of size effect, we do not have a general trend, we have trade-offs among our variables and **more negative correlation coefficients.**

The **highest correlation coefficient** is still between wider content choices and access to other countries' productions but after cleaning the data some negative correlations such as pop-ups and explicit advertisements and access to many languages in both audio and subtitles are even more negative.

We assume that: (1) the negative relationship among these variables is emphasized after eliminating the size effect, (2) people who are positively interested in wider content are also interested in access to other countries' production, and (3) as we have almost 0 correlation between the possibility to receive some malware or viruses while using illegal streaming platforms and feature to search for the preferred language there both of them has almost 0 effect on the other.

By looking at the eigenvalue table we can understand how each Principal Component is important in relation to the others.

	Eigenvalues of the Correlation Matrix			
	Eigenvalue	Difference	Proportion	Cumulative
1	4.74735791	1.38173574	0.2261	0.2261
2	3.36562218	1.32427774	0.1603	0.3863
3	2.04134444	0.49972166	0.0972	0.4835
4	1.54162278	0.31166480	0.0734	0.5569
5	1.22995798	0.20625344	0.0586	0.6155
6	1.02370454	0.17262394	0.0487	0.6643
7	0.85108061	0.05462133	0.0405	0.7048
8	0.79645927	0.04981793	0.0379	0.7427
9	0.74664134	0.09341625	0.0356	0.7783
10	0.65322509	0.03810037	0.0311	0.8094
11	0.61512471	0.06332176	0.0293	0.8387
12	0.55180295	0.03941626	0.0263	0.8649
13	0.51238669	0.09990237	0.0244	0.8893
14	0.41248432	0.02517186	0.0196	0.9090
15	0.38731245	0.04146380	0.0184	0.9274
16	0.34584865	0.03398025	0.0165	0.9439
17	0.31186840	0.04328843	0.0149	0.9588
18	0.26857997	0.02827287	0.0128	0.9715
19	0.24030711	0.04370289	0.0114	0.9830
20	0.19660422	0.03593985	0.0094	0.9923
21	0.16066437		0.0077	1.0000



Eigenvectors																						
		Prin1	Prin2	Prin3	Prin4	Prin5	Prin6	Prin7	Prin8	Prin9	Prin10	Prin11	Prin12	Prin13	Prin14	Prin15	Prin16	Prin17	Prin18	Prin19	Prin20	Prin21
new_q_1	Wide content	-.184529	0.228952	0.285510	-.182265	0.011341	-.119675	-.065983	0.060883	0.132146	0.539080	0.176092	-.279373	0.434362	-.274762	-.002011	0.246149	-.055406	0.143939	0.100779	-.021045	0.002950
new_q_2	Convenience	-.184338	0.212985	0.130889	-.309111	0.095585	0.375086	-.039262	-.381381	-.130220	-.237244	0.184481	-.283812	-.233814	-.016315	0.376132	0.129587	0.241923	0.033640	-.009391	0.175696	0.139689
new_q_3	Recommendations	-.182503	0.193677	0.328668	0.090906	0.147508	0.058632	-.334642	0.309069	0.281410	-.217662	0.304466	-.087640	-.010985	0.443154	-.039355	-.312332	-.113914	-.062247	0.140501	-.144940	-.005934
new_q_4	Customer service	-.206747	0.035086	0.343907	0.270252	0.124033	0.161495	-.270165	-.192466	0.072170	0.141274	-.001450	0.638606	-.066582	-.268061	0.022268	0.069905	0.152638	-.153606	-.222539	-.039727	-.023760
new_q_5	Quality	-.194034	0.061869	0.300350	-.191308	0.068432	-.569281	0.078114	-.208664	-.149412	0.017608	-.189185	0.186930	-.193884	0.417147	-.163440	0.175135	-.000169	0.199162	0.043326	0.183736	0.125102
new_q_6	Plans	-.236040	0.112209	0.221067	0.207188	0.180199	-.128420	0.236155	0.331129	-.063981	-.497871	-.317779	-.202707	0.135267	-.299438	-.088510	0.196075	0.250249	-.100530	-.025139	-.051551	-.034844
new_q_7	Languages	-.121920	0.288460	-.089406	-.054631	-.465390	0.123356	0.148592	-.126302	0.185365	-.232490	-.171249	0.281493	0.541100	0.207615	0.150454	-.075502	0.042142	0.172649	-.085078	-.034962	0.140931
new_q_8	Offline watching	-.135395	0.292845	0.145022	-.098274	-.268552	0.092448	0.354384	0.361960	0.060598	0.322320	-.228709	0.040711	-.429590	-.063673	0.207579	-.302234	-.041476	-.115203	-.044389	0.149732	-.063281
new_q_9	Cost	-.199678	0.001450	-.050914	0.283094	0.285049	0.309753	0.636250	-.036694	-.016316	0.094457	0.332349	0.103943	0.040007	0.217144	-.158184	0.185866	-.199171	0.108938	-.005067	0.054640	-.026745
new_q_10	Longer waiting	-.182431	-.156441	-.051394	0.420613	-.190193	-.454461	0.080234	-.065628	0.063951	-.052389	0.334618	-.098025	-.124717	-.133261	0.513459	-.074491	-.027671	0.194092	-.113262	-.122429	0.109853
new_q_11	Parental control	-.041711	0.050068	0.044944	0.584323	-.333256	0.220527	-.268736	-.017532	-.261447	0.188124	-.224524	-.258935	-.035101	0.247849	-.054647	0.217795	0.068644	0.058307	0.177601	0.214793	0.002370
new_q_12	Wider content	0.289806	0.314607	-.092650	0.079481	0.198754	-.085640	-.072578	0.086629	0.049937	0.046237	0.036952	-.049478	0.018051	0.167105	0.087451	0.039711	0.253876	0.319045	-.504780	0.145248	-.504443
new_q_13	Saving money	0.239970	0.316878	-.222849	0.051122	0.146647	-.112569	0.065567	0.111142	0.035551	0.129870	0.040089	0.212920	-.068347	0.155955	0.268347	0.299178	0.352116	-.203023	0.456469	-.319423	0.087847
new_q_14	Other countries	0.273681	0.281876	-.090061	0.172324	0.336279	0.002256	-.107456	0.098609	0.112013	-.037373	-.141841	0.047159	-.009728	-.178649	0.065457	-.058929	-.291389	0.217588	-.023897	0.336424	0.592808
new_q_15	Search language	0.256045	0.067732	0.299469	0.134048	0.179549	-.076980	0.179255	-.226621	-.475313	0.077284	0.007866	-.020705	0.327224	0.003070	0.142552	-.537064	0.077567	-.138696	0.150975	0.026471	-.059774
new_q_16	Pop-up	0.108717	-.384601	0.234861	-.062005	0.162637	0.178578	0.056077	0.155344	0.001772	0.203449	-.290655	-.125237	0.085615	0.308983	0.292076	0.107826	0.054875	0.023410	-.367373	-.349776	0.305746
new_q_17	Malware/viruses	0.063809	-.434413	0.162451	-.092117	0.036831	0.125074	0.003478	0.284184	0.099622	-.062343	0.019153	0.236653	0.087050	-.040832	0.237113	0.003324	0.208589	0.406930	0.404107	0.376476	-.157373
new_q_18	Downloading	0.317263	-.010136	0.238499	-.074853	-.256863	-.077746	0.023247	0.235730	-.171166	-.160979	0.321524	0.114661	0.108363	0.068373	0.146418	0.367305	-.233588	-.425286	-.189084	0.304118	0.021150
new_q_19	Friends	0.312949	0.063665	0.251010	-.003541	-.305297	0.080707	0.108719	0.101182	-.116508	-.041316	0.308227	0.024143	-.184866	-.132162	-.408317	-.049639	0.331166	0.343876	-.065801	-.242892	0.300447
new_q_20	Less waiting	0.307227	0.108577	0.328636	0.045768	-.063270	0.091171	0.064531	-.230388	0.120256	-.178129	-.205656	-.011368	-.165289	-.111979	0.129825	0.182172	-.485982	0.229971	0.196636	-.330773	-.313864
new_q_21	No content control	0.252107	-.122880	0.184988	0.150450	-.043815	-.075356	0.207830	-.327787	0.659191	0.039806	-.048325	-.213376	-.013451	0.053250	-.117242	-.035919	0.254097	-.277769	0.028872	0.244849	0.041376

Now the situation has changed: we can select the Principal Components 1-2-3-4-5-6 and also 7, since 0,85 is close to 1, thus representing 70% of the variability passing from a 21 dimensions space to a space with 7.

We can describe the other principal components as noise generated by the mind of the respondents.

We can prove that we have a 7-dimensional orthogonal space in which we can apply the **Euclidean distance** by demonstrating that correlation coefficients among the PCs are 0.

```
proc corr data=coord out=streaming3;
var Prin1-Prin7;
run;
```

The SAS System

The CORR Procedure

7 Variables:	Prin1 Prin2 Prin3 Prin4 Prin5 Prin6 Prin7

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
Prin1	102	0	2.17884	0	-6.08074	6.12942
Prin2	102	0	1.83456	0	-6.95854	3.38554
Prin3	102	0	1.42876	0	-5.04985	3.42753
Prin4	102	0	1.24162	0	-3.26875	3.43221
Prin5	102	0	1.10903	0	-2.13085	2.92813
Prin6	102	0	1.01178	0	-2.33365	3.10212
Prin7	102	0	0.92254	0	-2.83032	2.26200

Pearson Correlation Coefficients, N = 102
Prob > |r| under H0: Rho=0

	Prin1	Prin2	Prin3	Prin4	Prin5	Prin6	Prin7
Prin1	1.00000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000
Prin2	0.00000 1.0000	1.00000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000
Prin3	0.00000 1.0000	0.00000 1.0000	1.00000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000
Prin4	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	1.00000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000
Prin5	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	1.00000	0.00000 1.0000	0.00000 1.0000
Prin6	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	1.00000	0.00000 1.0000
Prin7	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	0.00000 1.0000	1.00000

Then we can proceed with the analysis of the **eigenvectors**, which show the corrrelation between variables and principal components

Eigenvectors

		Prin1	Prin2	Prin3	Prin4	Prin5	Prin6	Prin7
new_q_1	Wide content	-.184529	0.228952	0.285510	-.182265	0.011341	-.119675	-.065983
new_q_2	Convenience	-.184338	0.212985	0.130889	-.309111	0.095585	0.375086	-.039262
new_q_3	Recommendations	-.182503	0.193677	0.328668	0.090906	0.147508	0.058632	-.334642
new_q_4	Customer service	-.206747	0.035086	0.343907	0.270252	0.124033	0.161495	-.270165
new_q_5	Quality	-.194034	0.061869	0.300350	-.191308	0.068432	-.569281	0.078114
new_q_6	Plans	-.236040	0.112209	0.221067	0.207188	0.180199	-.128420	0.236155
new_q_7	Languages	-.121920	0.288460	-.089406	-.054631	-.465390	0.123356	0.148592
new_q_8	Offline watching	-.135395	0.292845	0.145022	-.098274	-.268552	0.092448	0.354384
new_q_9	Cost	-.199678	0.001450	-.050914	0.283094	0.285049	0.309753	0.636250
new_q_10	Longer waiting	-.182431	-.156441	-.051394	0.420613	-.190193	-.454461	0.080234
new_q_11	Parental control	-.041711	0.050068	0.044944	0.584323	-.333256	0.220527	-.268736
new_q_12	Wider content	0.289806	0.314607	-.092650	0.079481	0.198754	-.085640	-.072578
new_q_13	Saving money	0.239970	0.316878	-.222849	0.051122	0.146647	-.112569	0.065567
new_q_14	Other countries	0.273681	0.281876	-.090061	0.172324	0.336279	0.002256	-.107456
new_q_15	Search language	0.256045	0.067732	0.299469	0.134048	0.179549	-.076980	0.179255
new_q_16	Pop-up	0.108717	-.384601	0.234861	-.062005	0.162637	0.178578	0.056077
new_q_17	Malware/viruses	0.063809	-.434413	0.162451	-.092117	0.036831	0.125074	0.003478
new_q_18	Downloading	0.317263	-.010136	0.238499	-.074853	-.256863	-.077746	0.023247
new_q_19	Friends	0.312949	0.063665	0.251010	-.003541	-.305297	0.080707	0.108719
new_q_20	Less waiting	0.307227	0.108577	0.328636	0.045768	-.063270	0.091171	0.064531
new_q_21	No content control	0.252107	-.122880	0.184988	0.150450	-.043815	-.075356	0.207830

PRINCIPAL COMPONENT 1

This principal component is positively affected mainly by less waiting time, no content control, the possibility to watch with friends, wider content, and preferred search language. However it is negatively affected mainly by longer waiting time, customer service, parental control, subscription plans, and quality. We could see the opposition of interest between the cost and wide spectrum of choices.

PRINCIPAL COMPONENT 2

Here we have an opposition between the possibility to watch offline on Netflix, different language selection, and other countries' production that have a positive correlation with the second principal component and longer waiting time due to copyright and the possibility of receiving malware or viruses. This might represent the part of the population that is more interested in watching convenience and wide content rather than the side effects that could be caused or appear.

PRINCIPAL COMPONENT 3

This principal component is mainly positively affected by customer service, recommendations generated by Netflix, and the quality of the watching (High resolution, no ads). On the negative side, we can find saving money and the cost of Netflix subscription. This represents the part that is willing to pay more to get better quality.

PRINCIPAL COMPONENT 4

This is positively affected by customer service, parental control, and longer waiting time due to copyrights, while being negatively affected by convenience, wide content, and quality. This might indicate an opposition between those people who care more about humanistic values rather than the quality of watching or huge choices.

PRINCIPAL COMPONENT 5

Here we have an opposition between cost, subscription plans, and other countries' production that are positively correlated with the PC in question, while offline watching on Netflix and downloading on illegal platforms are negatively correlated.

PRINCIPAL COMPONENT 6

This principal component might indicate those people who show a positive interest in convenience, cost, and parental control but a negative interest in the quality and longer waiting time.

PRINCIPAL COMPONENT 7

Here we can see the part of the people who are positively affected by offline watching and cost and mainly negatively by recommendations and parental control.

3.4 CLUSTERING METHOD AND FORMATION

Through SAS calculations we have so far created an *orthogonal space*, then proceed with the clusterization process.

Clustering is the task of grouping a set of objects in such a way that the units in the same group, i.e. "*cluster*", are similar to each other and different from those in other clusters. By profiling individuals using the clusterization process we can better study patterns of behavior among them.

Different clustering methods can be used. One of these is called the "*minimum linkage method*" and it consists in calculating the minimum distances among units with an algorithm that uses the Euclidean distance as a measure. This method results in an approximated classification scheme since it only uses the major characteristics of the units to form homogeneous groups. This causes the generation of errors as their heterogeneity increases.

Another method called the "*Ward algorithm*" is the method we decided to employ, mainly because it is useful to overcome issues deriving from nested dendrograms (hierarchical clusters). A *nested dendrogram* is a cluster tree used to represent data, where each group links to two or more successor groups and, ideally, ends up showing a meaningful classification scheme. Unfortunately, that is rarely the case with surveys like ours, where opinions are asked to individuals. People tend, in fact, to avoid extreme positions when expressing a point of view and this causes a fallacy in the minimum linkage method which fails to recognize natural groups among the data.
The *Ward algorithm* focuses on minimizing inter-cluster variability and maximizing external variability, hence maximizing the diversity among different groups. It is this opposite approach compared to the minimum linkage method that allows us to overcome the problem of hierarchical clusters and analyze the "*fusion distance, (FD)*" i.e. the increase in internal variability as groups are unified with each other.

3.5 DENDROGRAM AND IDENTIFICATION

To create clusters using the Ward's method we used the command:

```
proc cluster data=coord method=ward outtree=tree;
var Prin1-Prin7;
id id;
run;
```

Which gaver us a useful insight of the clusters' analysis:

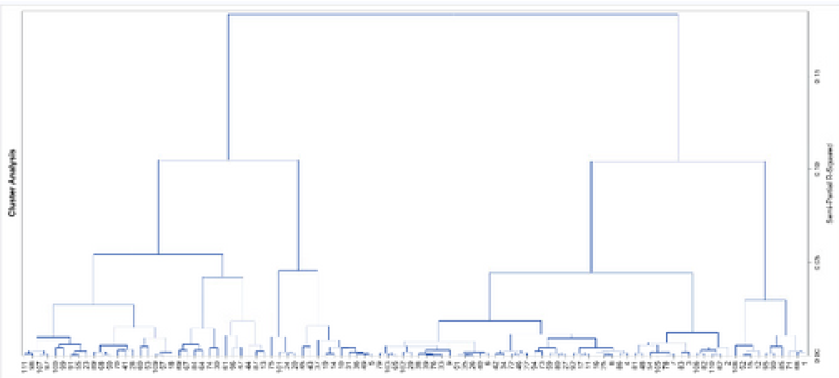
The SAS System

The CLUSTER Procedure
Ward's Minimum Variance Cluster Analysis

Eigenvalues of the Covariance Matrix				
	Eigenvalue	Difference	Proportion	Cumulative
1	4.74735791	1.38173574	0.3208	0.3208
2	3.36562218	1.32427774	0.2274	0.5481
3	2.04134444	0.49972166	0.1379	0.6861
4	1.54162278	0.31166480	0.1042	0.7902
5	1.22995798	0.20625344	0.0831	0.8733
6	1.02370454	0.17262394	0.0692	0.9425
7	0.85108061		0.0575	1.0000

Root-Mean-Square Total-Sample Standard Deviation	1.454092

Root-Mean-Square Distance Between Observations	5.440715

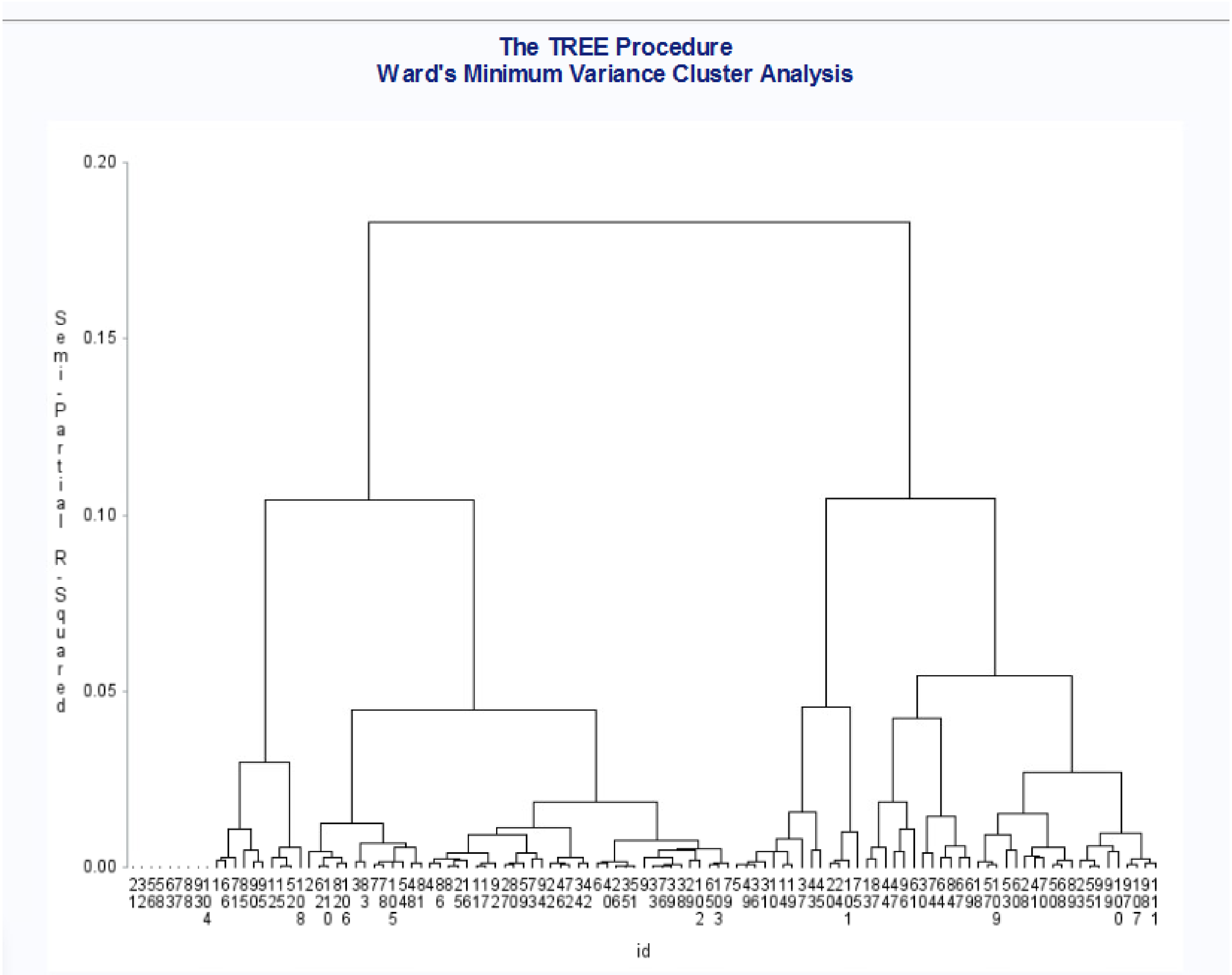


Cluster History						
Number of Clusters	Clusters Joined		Freq	Semipartial R-Square	R-Square	Tie
101	26	35	2	0.0002	1.00	
100	33	76	2	0.0003	1.00	
99	55	91	2	0.0004	.999	
98	8	25	2	0.0004	.999	
97	15	52	2	0.0004	.998	
96	CL100	39	3	0.0005	.998	
95	11	17	2	0.0005	.997	
94	CL101	51	3	0.0006	.997	
93	62	110	2	0.0006	.996	
92	46	72	2	0.0006	.996	
91	50	68	2	0.0006	.995	
90	5	49	2	0.0007	.994	
89	97	107	2	0.0007	.994	
88	65	103	2	0.0007	.993	
87	57	109	2	0.0007	.992	
86	7	78	2	0.0008	.991	
85	27	80	2	0.0008	.991	
84	14	19	2	0.0009	.990	
83	CL90	36	3	0.0009	.989	
82	CL96	38	4	0.0010	.988	
81	98	111	2	0.0010	.987	
80	34	42	2	0.0010	.986	
79	48	81	2	0.0011	.985	
78	CL85	59	3	0.0012	.984	
77	4	86	2	0.0012	.982	
76	20	24	2	0.0012	.981	
75	82	106	2	0.0012	.980	
74	CL95	92	3	0.0013	.979	
73	CL88	79	3	0.0013	.977	
72	22	CL92	3	0.0013	.976	
71	6	40	2	0.0013	.975	
70	90	95	2	0.0014	.973	
69	CL86	105	3	0.0015	.972	
68	3	83	2	0.0016	.970	
67	CL71	CL94	5	0.0016	.969	
66	CL69	54	4	0.0017	.967	
65	18	CL87	3	0.0017	.965	
64	CL83	31	4	0.0017	.963	
63	CL98	16	3	0.0018	.962	
62	1	66	2	0.0019	.960	
61	CL91	89	3	0.0020	.958	
60	CL76	101	3	0.0020	.956	
59	29	102	2	0.0021	.954	
58	23	CL99	3	0.0021	.952	
57	73	94	2	0.0022	.949	
56	13	87	2	0.0022	.947	
55	CL77	CL63	5	0.0024	.945	
54	CL89	CL81	4	0.0025	.942	
53	CL62	71	3	0.0026	.940	
52	12	CL97	3	0.0026	.937	
51	CL72	CL80	5	0.0027	.934	
50	41	70	2	0.0027	.932	
49	67	69	2	0.0028	.929	
48	64	84	2	0.0028	.926	
47	CL93	CL75	4	0.0028	.923	
46	9	CL82	5	0.0029	.920	
45	28	CL50	3	0.0033	.917	
44	CL78	CL57	5	0.0040	.913	
43	30	74	2	0.0040	.909	
42	CL55	CL74	8	0.0041	.905	
41	CL64	10	5	0.0042	.901	
40	99	100	2	0.0042	.896	
39	47	96	2	0.0043	.892	
38	2	CL47	5	0.0045	.888	
37	CL41	CL84	7	0.0045	.883	
36	53	60	2	0.0046	.878	
35	43	45	2	0.0048	.874	
34	85	CL70	3	0.0049	.869	
33	CL46	CL59	7	0.0049	.864	
32	CL33	CL73	10	0.0050	.859	
31	CL45	CL61	6	0.0054	.853	
30	CL52	108	4	0.0055	.848	
29	CL56	44	3	0.0057	.842	
28	CL66	CL79	6	0.0058	.836	
27	CL58	CL40	5	0.0060	.830	
26	CL48	CL49	4	0.0061	.824	
25	CL68	CL28	8	0.0070	.817	
24	CL67	CL32	15	0.0076	.810	
23	CL37	37	8	0.0081	.802	
22	CL42	CL44	13	0.0092	.792	
21	CL65	CL36	5	0.0093	.783	
20	CL27	CL54	9	0.0097	.773	
19	CL60	75	4	0.0101	.763	
18	CL53	CL34	6	0.0107	.753	
17	CL39	61	3	0.0109	.742	
16	CL22	CL51	18	0.0112	.730	
15	CL38	CL25	13	0.0123	.718	
14	CL43	CL26	6	0.0145	.704	
13	CL21	CL31	11	0.0153	.688	
12	CL23	CL35	10	0.0157	.673	
11	CL29	CL17	6	0.0185	.654	
10	CL16	CL24	33	0.0185	.636	
9	CL13	CL20	20	0.0271	.609	
8	CL18	CL30	10	0.0297	.579	
7	CL11	CL14	12	0.0422	.537	
6	CL15	CL10	46	0.0447	.492	
5	CL12	CL19	14	0.0454	.447	
4	CL7	CL9	32	0.0544	.392	
3	CL8	CL6	56	0.1041	.288	
2	CL5	CL4	46	0.1048	.183	
1	CL3	CL2	102	0.1832	.000	

Followed by:

```
proc tree data=tree;
id id;
run;
```

The combination of these two commands resulted into this graph:



The next step is deciding how many clusters to select. In doing so, trade-offs between the reduction of clusters' complexity and the increase of Fusion Distance (internal variability due to merging groups together) must be carefully taken into consideration.
We decided that for the purpose of our research, 5 clusters would work best, hence why we used this command on SAS:

```
proc tree data=tree out=clus nclusters=5;
id id;
run;
```

What this formula does is reorganizing data into clusters so to match our desire of only keeping 5 of them.

We then proceeded with running a frequency analysis in order to understand how many respondents we had at this point in each cluster. Additionally, we performed the *proc means* formula which allowed us to acknowledge which variables characterized the new clusters.

```
proc sort data=clus;by id;run;
proc sort data=sample0;by id;run;


data streaming_cluster;
MERGE clus streaming2;
by id;
run;
proc means data=streaming_cluster;
var new_q_1-new_q_21;
class cluster;
run;
```

The MEANS Procedure

CLUSTER	N Obs	Variable	Label	N	Mean	Std Dev	Minimum	Maximum
1	46	new_q_1	Wide content	46	0.4293686	0.3959165	-0.2758621	1.0000000
		new_q_2	Convenience	46	0.6211913	0.5022138	-1.0000000	1.0000000
		new_q_3	Recommendations	46	0.2533390	0.5420074	-1.0000000	1.0000000
		new_q_4	Customer service	46	-0.0654989	0.4374936	-1.0000000	1.0000000
		new_q_5	Quality	46	0.3803916	0.4776805	-0.7613636	1.0000000
		new_q_6	Plans	46	0.2178142	0.4588527	-0.4166667	1.0000000
		new_q_7	Languages	46	0.6577898	0.4916881	-0.7042254	1.0000000
		new_q_8	Offline watching	46	0.6310086	0.4459631	-0.2840909	1.0000000
		new_q_9	Cost	46	-0.2445860	0.5865506	-1.0000000	1.0000000
		new_q_10	Longer waiting	46	-0.4022838	0.3483702	-1.0000000	0.2758621
		new_q_11	Parental control	46	-0.0956878	0.4160556	-1.0000000	1.0000000
		new_q_12	Wider content	46	0.8343684	0.3135548	-0.1923077	1.0000000
		new_q_13	Saving money	46	0.7847339	0.3540728	-0.2222222	1.0000000
		new_q_14	Other countries	46	0.7370176	0.3864193	-0.1710526	1.0000000
		new_q_15	Search language	46	0.3692976	0.5723396	-0.7586207	1.0000000
		new_q_16	Pop-up	46	-0.9102333	0.2171357	-1.0000000	-0.0869565
		new_q_17	Malware/viruses	46	-0.9544445	0.1755990	-1.0000000	-0.1923077
		new_q_18	Downloading	46	0.1657969	0.5927267	-1.0000000	1.0000000
		new_q_19	Friends	46	0.3079748	0.5851106	-1.0000000	1.0000000
		new_q_20	Less waiting	46	0.3322109	0.5379942	-0.7042254	1.0000000
		new_q_21	No content control	46	-0.1218746	0.5357989	-1.0000000	1.0000000
2	20	new_q_1	Wide content	20	0.5259507	0.6193731	-1.0000000	1.0000000
		new_q_2	Convenience	20	0.7784784	0.3386067	-0.0222222	1.0000000
		new_q_3	Recommendations	20	0.2492268	0.7298768	-1.0000000	1.0000000
		new_q_4	Customer service	20	0.0867736	0.7744343	-1.0000000	1.0000000
		new_q_5	Quality	20	0.3649799	0.6295541	-1.0000000	1.0000000
		new_q_6	Plans	20	0.1075837	0.6563941	-1.0000000	1.0000000
		new_q_7	Languages	20	0.4777646	0.6073240	-0.5961538	1.0000000
		new_q_8	Offline watching	20	0.6141463	0.6090982	-1.0000000	1.0000000
		new_q_9	Cost	20	-0.1755649	0.8108994	-1.0000000	1.0000000
		new_q_10	Longer waiting	20	-0.5211927	0.5050955	-1.0000000	1.0000000
		new_q_11	Parental control	20	-0.4760965	0.4876444	-1.0000000	0.4615385
		new_q_12	Wider content	20	-0.0561190	0.2135616	-0.6818182	0.2881356
		new_q_13	Saving money	20	-0.0716422	0.4428734	-1.0000000	1.0000000
		new_q_14	Other countries	20	0.0658751	0.1953585	-0.2075472	0.5961538
		new_q_15	Search language	20	-0.0036236	0.2830691	-0.4683544	1.0000000
		new_q_16	Pop-up	20	-0.0983987	0.4121299	-1.0000000	1.0000000
		new_q_17	Malware/viruses	20	-0.0939711	0.4604734	-1.0000000	1.0000000
		new_q_18	Downloading	20	0.0364801	0.2942130	-0.7341772	0.7162162
		new_q_19	Friends	20	0.0148795	0.2520724	-0.4683544	0.7162162
		new_q_20	Less waiting	20	-0.0139887	0.3461488	-1.0000000	0.5961538
		new_q_21	No content control	20	-0.1126684	0.3988767	-1.0000000	1.0000000
3	10	new_q_1	Wide content	10	-0.4040936	0.3647932	-1.0000000	0.0156250
		new_q_2	Convenience	10	-0.2647689	0.5659497	-1.0000000	1.0000000
		new_q_3	Recommendations	10	-0.6113507	0.4212328	-1.0000000	-0.0232558
		new_q_4	Customer service	10	-0.5226751	0.3026814	-1.0000000	-0.1710526
		new_q_5	Quality	10	-0.6710124	0.5179801	-1.0000000	0.5333333
		new_q_6	Plans	10	-0.6899914	0.2067422	-1.0000000	-0.3000000
		new_q_7	Languages	10	0.1584168	0.7511459	-1.0000000	1.0000000
		new_q_8	Offline watching	10	0.0524375	0.6097658	-1.0000000	1.0000000
		new_q_9	Cost	10	-0.5789057	0.5606641	-1.0000000	0.6500000
		new_q_10	Longer waiting	10	-0.5526804	0.4237632	-1.0000000	0.0307692
		new_q_11	Parental control	10	-0.0160585	0.5119075	-0.6500000	1.0000000
		new_q_12	Wider content	10	0.9676923	0.1021659	0.6769231	1.0000000
		new_q_13	Saving money	10	1.0000000	0	1.0000000	1.0000000
		new_q_14	Other countries	10	0.9676923	0.1021659	0.6769231	1.0000000
		new_q_15	Search language	10	0.3753366	0.6856689	-0.6612903	1.0000000
		new_q_16	Pop-up	10	-0.1228817	0.8339132	-1.0000000	1.0000000
		new_q_17	Malware/viruses	10	-0.4169286	0.6486372	-1.0000000	1.0000000
		new_q_18	Downloading	10	0.6642558	0.5904673	-0.6557377	1.0000000
		new_q_19	Friends	10	0.6374519	0.6712891	-1.0000000	1.0000000
		new_q_20	Less waiting	10	0.8360400	0.4161737	-0.3114754	1.0000000
		new_q_21	No content control	10	0.6823798	0.6217220	-1.0000000	1.0000000
4	14	new_q_1	Wide content	14	0.4517766	0.5755225	-1.0000000	1.0000000
		new_q_2	Convenience	14	0.6278346	0.4127539	-0.3731343	1.0000000
		new_q_3	Recommendations	14	0.2931722	0.5656654	-1.0000000	0.7341772
		new_q_4	Customer service	14	0.1795149	0.6491838	-1.0000000	1.0000000
		new_q_5	Quality	14	0.2339005	0.7211061	-1.0000000	1.0000000
		new_q_6	Plans	14	0.3806157	0.3670622	-0.3000000	1.0000000
		new_q_7	Languages	14	0.7435689	0.4725956	-0.2500000	1.0000000
		new_q_8	Offline watching	14	0.4484151	0.5705430	-1.0000000	1.0000000
		new_q_9	Cost	14	0.0093363	0.5487872	-1.0000000	0.7000000
		new_q_10	Longer waiting	14	-0.0537814	0.4003618	-1.0000000	0.4683544
		new_q_11	Parental control	14	-0.0110389	0.6479758	-1.0000000	1.0000000
		new_q_12	Wider content	14	0.0385157	0.8905141	-1.0000000	1.0000000
		new_q_13	Saving money	14	0.1915689	0.8288100	-1.0000000	1.0000000
		new_q_14	Other countries	14	0.2606646	0.8649411	-1.0000000	1.0000000
		new_q_15	Search language	14	-0.7694759	0.4323958	-1.0000000	0.3114754
		new_q_16	Pop-up	14	-1.0000000	0	-1.0000000	-1.0000000
		new_q_17	Malware/viruses	14	-0.9423077	0.2158648	-1.0000000	-0.1923077
		new_q_18	Downloading	14	-0.8388736	0.3572229	-1.0000000	0.1250000
		new_q_19	Friends	14	-0.9769231	0.0863459	-1.0000000	-0.6769231
		new_q_20	Less waiting	14	-0.8866820	0.2937689	-1.0000000	-0.0597015
		new_q_21	No content control	14	-0.9391892	0.2275332	-1.0000000	-0.1436486
5	12	new_q_1	Wide content	12	-0.0332647	0.8526615	-1.0000000	1.0000000
		new_q_2	Convenience	12	0.1181291	0.7785017	-1.0000000	1.0000000
		new_q_3	Recommendations	12	0.3133765	0.7892823	-1.0000000	1.0000000
		new_q_4	Customer service	12	0.4779804	0.5794941	-0.3125000	1.0000000
		new_q_5	Quality	12	0.2692616	0.6742848	-1.0000000	1.0000000
		new_q_6	Plans	12	0.6902699	0.4037616	-0.0232558	1.0000000
		new_q_7	Languages	12	0.0429515	0.7850008	-1.0000000	1.0000000
		new_q_8	Offline watching	12	-0.0607063	0.8359432	-1.0000000	1.0000000
		new_q_9	Cost	12	0.3205304	0.7712802	-1.0000000	1.0000000
		new_q_10	Longer waiting	12	0.1867122	0.6091899	-0.7613636	1.0000000
		new_q_11	Parental control	12	0.4565343	0.6079037	-0.3823529	1.0000000
		new_q_12	Wider content	12	0.0264951	0.6369635	-1.0000000	1.0000000
		new_q_13	Saving money	12	-0.2435200	0.4877090	-1.0000000	0.3823529
		new_q_14	Other countries	12	0.2369836	0.5886068	-1.0000000	1.0000000
		new_q_15	Search language	12	0.2772019	0.5115829	-0.5434783	1.0000000
		new_q_16	Pop-up	12	0.0239444	0.7532182	-1.0000000	1.0000000
		new_q_17	Malware/viruses	12	-0.1778448	0.7251742	-1.0000000	1.0000000
		new_q_18	Downloading	12	-0.0421302	0.5367765	-0.7613636	1.0000000
		new_q_19	Friends	12	0.0951295	0.4523976	-0.3152174	1.0000000
		new_q_20	Less waiting	12	0.2318001	0.5937524	-0.6181818	1.0000000
		new_q_21	No content control	12	0.1705725	0.5671262	-0.7613636	1.0000000

Clusters are strongly positively characterized by those variables that have a positive mean and negatively characterized by those variables with a negative mean.

3.6 T-TEST

In order to understand if the difference between the two means is real or due to random effect, performing a **T-Test analysis** was our next move. This analysis is useful to recognize if the relationship between quantitative variables and clusters is noteworthy.
The study consists in examining if, for each variable, the cluster average is different from the sample average *only* because of random effect.
We therefore created two main hypotheses that we then set against each other: H0 is called the "null hypothesis" and states that the cluster average and the sample average are equal; H1, called the "alternative hypothesis", simply states that the two of them differ.
A *p-value*, i.e. the probability of observing our H0 hypothesis to be true, was then needed to be found. Any p-value is connected to a *t-value,* defined as the ratio of the difference between the cluster mean and the population mean, and the standard error. Usually, we reject H0 for p-values that are lower than 0.05 and then state that the variable is able to differentiate the cluster.

T-values can be estimated using one of two methods **Satterthwaite method** and **Pooled method**. Since we had unequal variances we used the first one.
To run the t-test we generated a sixth 'fake' cluster containing all the units in the data set, i.e. representing the general average of the sample so that it could be used as a comparison between the single clusters and the general sample.

```
data streaming_tricked;
set streaming_cluster;
cluster=6;
run;

data streaming_app;
set streaming_cluster streaming_tricked;
run;
```

To analyze each cluster we used the commands below where each code describes each cluster.

```
proc ttest data=streaming_app;
var new_q_1-new_q_21;
where cluster=1 or cluster=6;
class cluster;
ods output ttests=cluster_1;
run;

proc ttest data=streaming_app;
var new_q_1-new_q_21;
where cluster=2 or cluster=6;
class cluster;
ods output ttests=cluster_2;
run;

proc ttest data=streaming_app;
var new_q_1-new_q_21;
where cluster=3 or cluster=6;
class cluster;
ods output ttests=cluster_3;
run;

proc ttest data=streaming_app;
var new_q_1-new_q_21;
where cluster=4 or cluster=6;
class cluster;
ods output ttests=cluster_4;
run;

proc ttest data=streaming_app;
var new_q_1-new_q_21;
where cluster=5 or cluster=6;
class cluster;
ods output ttests=cluster_5;
run;
```

```
data cluster_1_1; set cluster_1;
where variances="Unequal";
rename tvalue=tvalue_cl_1;
rename probt=probt_cl_1;
run;

data cluster_2_1; set cluster_2;
where variances="Unequal";
rename tvalue=tvalue_cl_2;
rename probt=probt_cl_2;
run;

data cluster_3_1; set cluster_3;
where variances="Unequal";
rename tvalue=tvalue_cl_3;
rename probt=probt_cl_3;
run;

data cluster_4_1; set cluster_4;
where variances="Unequal";
rename tvalue=tvalue_cl_4;
rename probt=probt_cl_4;
run;

data cluster_5_1; set cluster_5;
where variances="Unequal";
rename tvalue=tvalue_cl_5;
rename probt=probt_cl_5;
run;
```

Additional data on the single clusters' analysis below:

The TTEST Procedure

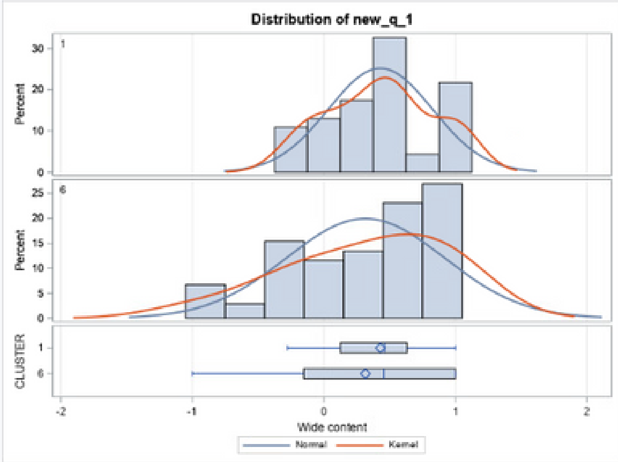
Variable: new_q_1 (Wide content)

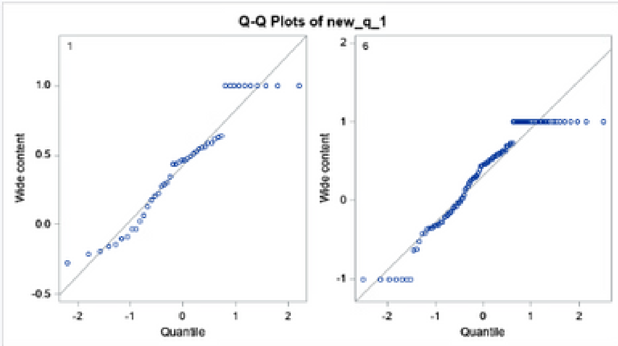
CLUSTER	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
1		46	0.4294	0.3959	0.0584	-0.2759	1.0000
6		104	0.3156	0.6003	0.0589	-1.0000	1.0000
Diff (1-2)	Pooled		0.1138	0.5463	0.0967		
Diff (1-2)	Satterthwaite		0.1138		0.0829		

CLUSTER	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
1		0.4294	0.3118	0.5469	0.3959	0.3284	0.4987
6		0.3156	0.1989	0.4323	0.6003	0.5283	0.6951
Diff (1-2)	Pooled	0.1138	-0.0774	0.3049	0.5463	0.4905	0.6165
Diff (1-2)	Satterthwaite	0.1138	-0.0503	0.2778			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	148	1.18	0.2414
Satterthwaite	Unequal	126.08	1.37	0.1723

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	103	45	2.30	0.0023





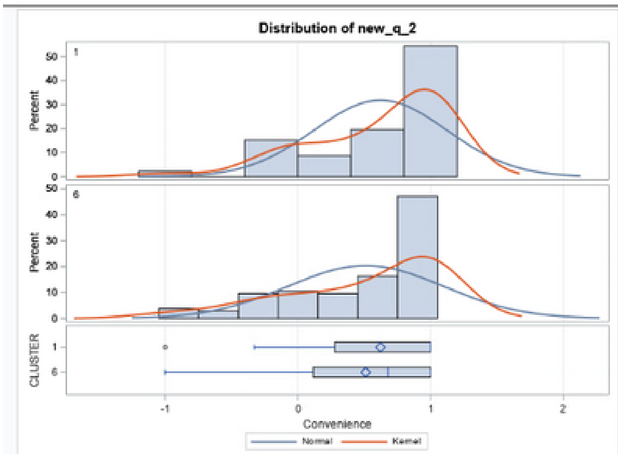
Variable: new_q_2 (Convenience)

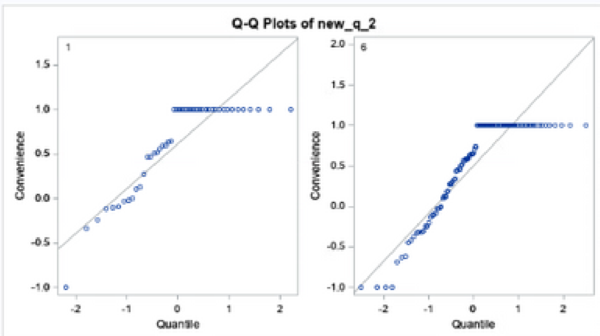
CLUSTER	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
1		46	0.6212	0.5022	0.0740	-1.0000	1.0000
6		104	0.5100	0.5880	0.0577	-1.0000	1.0000
Diff (1-2)	Pooled		0.1112	0.5633	0.0997		
Diff (1-2)	Satterthwaite		0.1112		0.0938		

CLUSTER	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
1		0.6212	0.4721	0.7703	0.5022	0.4166	0.6325
6		0.5100	0.3956	0.6243	0.5880	0.5175	0.6809
Diff (1-2)	Pooled	0.1112	-0.0859	0.3083	0.5633	0.5058	0.6357
Diff (1-2)	Satterthwaite	0.1112	-0.0750	0.2974			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	148	1.11	0.2667
Satterthwaite	Unequal	100.05	1.19	0.2388

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	103	45	1.37	0.2366





3.7 CLUSTER ANALYSIS

3.7.I CLUSTER 1

	Variable	Method	Variances	t Value	DF	Pr > |t|
1	new_q_1	Satterthwaite	Unequal	1.37	126.080	0.1723
2	new_q_2	Satterthwaite	Unequal	1.19	100.050	0.2388
3	new_q_3	Satterthwaite	Unequal	0.61	102.580	0.5452
4	new_q_4	Satterthwaite	Unequal	-0.9	115.770	0.3703
5	new_q_5	Satterthwaite	Unequal	1.31	113.660	0.1934
6	new_q_6	Satterthwaite	Unequal	0.5	108.460	0.6180
7	new_q_7	Satterthwaite	Unequal	1.86	108.980	0.0658
8	new_q_8	Satterthwaite	Unequal	1.78	115.280	0.0782
9	new_q_9	Satterthwaite	Unequal	-0.74	99.505	0.4639
10	new_q_10	Satterthwaite	Unequal	-1.38	120.920	0.1708
11	new_q_11	Satterthwaite	Unequal	-0.03	115.540	0.9737
12	new_q_12	Satterthwaite	Unequal	4.91	147.300	<.0001
13	new_q_13	Satterthwaite	Unequal	4.22	140.570	<.0001
14	new_q_14	Satterthwaite	Unequal	2.95	120.080	0.0038
15	new_q_15	Satterthwaite	Unequal	2.29	95.221	0.0239
16	new_q_16	Satterthwaite	Unequal	-5.14	146.980	<.0001
17	new_q_17	Satterthwaite	Unequal	-5.44	143.620	<.0001
18	new_q_18	Satterthwaite	Unequal	1.26	91.826	0.2117
19	new_q_19	Satterthwaite	Unequal	2.13	96.417	0.0355
20	new_q_20	Satterthwaite	Unequal	1.85	103.430	0.0667
21	new_q_21	Satterthwaite	Unequal	0.06	100.160	0.9490

In this case we can observe that our p-values resulted lower than 0.05 for our new_q_12, new_q_13, new_q_14, new_q_15, new_q_16, new_q_17 and new_q_19 variables (*wider content, saving money, other countries, search language, pop-up, malware/viruses and friends* respectively). This means that these variables are of particular importance for the differentiation process of the cluster we are analysing here.
On the contrary, we found our new_q_11 and new_q_21 *variables (Parental control and No content control* respectively*)* having particularly large p-values, hence not being statistically significant.
These results additionally show that people do not particularly care about *Customer service, cost,* and *longer waiting time (*new_q_4, new_q_9, new_q_10) when it comes to Netflix usage characteristics, but in contrast, it shows that people **strongly dislike** (highly negative t-values) *pop-ups* and *the possibility of catching viruses (*new_q_16, new_q_17*)* on illegal streaming platforms.
Safe to say that if any of our respondents uses illegal streaming platforms, it probably won't be because of these last two features, but it will most likely have to do with a *wider content choice* compared to Netflix (*new_q_12*) since its t-value is 4.91, showing a **strong association** to the choice, and also with *saving money (new_q_13)* with t-value of 4.22.

3.7.II CLUSTER 2

	Variable	Method	Variances	t Value	DF	Pr > |t|
1	new_q_1	Satterthwaite	Unequal	1.40	26.325	0.1738
2	new_q_2	Satterthwaite	Unequal	2.82	44.658	0.0071
3	new_q_3	Satterthwaite	Unequal	0.33	25.158	0.7435
4	new_q_4	Satterthwaite	Unequal	0.40	23.588	0.6905
5	new_q_5	Satterthwaite	Unequal	0.70	27.050	0.4900
6	new_q_6	Satterthwaite	Unequal	-0.42	25.116	0.6793
7	new_q_7	Satterthwaite	Unequal	-0.02	27.397	0.9827
8	new_q_8	Satterthwaite	Unequal	0.95	26.749	0.3524
9	new_q_9	Satterthwaite	Unequal	-0.06	24.437	0.9536
10	new_q_10	Satterthwaite	Unequal	-1.76	26.662	0.0898
11	new_q_11	Satterthwaite	Unequal	-3.13	29.791	0.0039
12	new_q_12	Satterthwaite	Unequal	-6.60	89.451	<.0001
13	new_q_13	Satterthwaite	Unequal	-4.40	35.838	<.0001
14	new_q_14	Satterthwaite	Unequal	-6.33	85.007	<.0001
15	new_q_15	Satterthwaite	Unequal	-1.48	62.922	0.1427
16	new_q_16	Satterthwaite	Unequal	4.34	35.935	0.0001
17	new_q_17	Satterthwaite	Unequal	4.68	29.899	<.0001
18	new_q_18	Satterthwaite	Unequal	0.06	59.371	0.9514
19	new_q_19	Satterthwaite	Unequal	-0.74	77.302	0.4610
20	new_q_20	Satterthwaite	Unequal	-1.57	49.481	0.1230
21	new_q_21	Satterthwaite	Unequal	0.14	39.724	0.8860

The variables characterizing this cluster are new_q_2 (*Convenience*), new_q_11 (*Parental control*), new_q_12 (*Wider content*), new_q_13 (*Saving money*), new_q_14 (*Access to other countries' production*), new_q_16 (*Pop-ups*), and new_q_17 (*Malware/viruses*).
As we can see from their **negative** t-values though, individuals in this cluster do not use Netflix for parental control nor do they use Illegal streaming platforms for wider content, saving money purposes or other countries' production.

3.7.III CLUSTER 3

	Variable	Method	Variances	t Value	DF	Pr > |t|
1	new_q_1	Satterthwaite	Unequal	-5.56	14.212	<.0001
2	new_q_2	Satterthwaite	Unequal	-4.12	10.955	0.0017
3	new_q_3	Satterthwaite	Unequal	-5.43	13.553	<.0001
4	new_q_4	Satterthwaite	Unequal	-4.77	16.886	0.0002
5	new_q_5	Satterthwaite	Unequal	-5.29	11.794	0.0002
6	new_q_6	Satterthwaite	Unequal	-9.94	26.737	<.0001
7	new_q_7	Satterthwaite	Unequal	-1.31	10.245	0.2172
8	new_q_8	Satterthwaite	Unequal	-2.09	10.783	0.0615
9	new_q_9	Satterthwaite	Unequal	-2.19	11.722	0.0496
10	new_q_10	Satterthwaite	Unequal	-1.74	11.553	0.1085
11	new_q_11	Satterthwaite	Unequal	0.45	11.234	0.6618
12	new_q_12	Satterthwaite	Unequal	7.36	90.336	<.0001
13	new_q_13	Satterthwaite	Unequal	8.99	104.000	<.0001
14	new_q_14	Satterthwaite	Unequal	7.35	77.198	<.0001
15	new_q_15	Satterthwaite	Unequal	1.09	10.550	0.2981
16	new_q_16	Satterthwaite	Unequal	1.66	9.886	0.1289
17	new_q_17	Satterthwaite	Unequal	1.03	10.224	0.3269
18	new_q_18	Satterthwaite	Unequal	3.22	11.091	0.0081
19	new_q_19	Satterthwaite	Unequal	2.52	10.729	0.0289
20	new_q_20	Satterthwaite	Unequal	4.73	13.671	0.0003
21	new_q_21	Satterthwaite	Unequal	3.94	10.839	0.0024

Several variables are noteworthy here. All variables between new_q_1 and new_q_6 (i.e. *Wide content, Convenience, Recommendations, Customer service, Quality, Plans*), along with variables new-q-12,_13,_14,_18,_20 and_21 (*Wider content, Saving money, Other countries' production, Downloading, Less waiting time*) seem to characterize it.
The **negatively strong relation** with variable new_q_6 (*Plans*) and the **positively strong one** with variable new_q_13 (*Saving money*) are the ones better-describing similarities between individuals in this cluster.

3.7.IV CLUSTER 4

	Variable	Method	Variances	t Value	DF	Pr > |t|
1	new_q_1	Satterthwaite	Unequal	0.83	17.040	0.4197
2	new_q_2	Satterthwaite	Unequal	0.95	20.877	0.3545
3	new_q_3	Satterthwaite	Unequal	0.62	17.984	0.5424
4	new_q_4	Satterthwaite	Unequal	0.91	16.139	0.3770
5	new_q_5	Satterthwaite	Unequal	-0.11	15.860	0.9100
6	new_q_6	Satterthwaite	Unequal	1.82	23.030	0.0812
7	new_q_7	Satterthwaite	Unequal	1.87	19.768	0.0766
8	new_q_8	Satterthwaite	Unequal	-0.15	17.201	0.8800
9	new_q_9	Satterthwaite	Unequal	1.08	18.864	0.2953
10	new_q_10	Satterthwaite	Unequal	2.13	18.927	0.0465
11	new_q_11	Satterthwaite	Unequal	1.29	15.796	0.6586
12	new_q_12	Satterthwaite	Unequal	-1.7	14.779	0.1094
13	new_q_13	Satterthwaite	Unequal	-1.09	15.109	0.2931
14	new_q_14	Satterthwaite	Unequal	-1.03	14.449	0.3197
15	new_q_15	Satterthwaite	Unequal	-6.83	21.474	<.0001
16	new_q_16	Satterthwaite	Unequal	-7.42	105.000	<.0001
17	new_q_17	Satterthwaite	Unequal	-3.93	40.325	0.0003
18	new_q_18	Satterthwaite	Unequal	-7.64	25.763	<.0001
19	new_q_19	Satterthwaite	Unequal	-15.41	117.990	<.0001
20	new_q_20	Satterthwaite	Unequal	-10.18	34.028	<.0001
21	new_q_21	Satterthwaite	Unequal	-9.38	47.044	<.0001

Important variables mingled with this cluster's units are from variable new_q_14 to new_q_21 (*Other countries' production, Search language, Pop-up, Malware/viruses, Downloading, Friends, Less waiting time, and No content control* respectively) and they all **strongly influence** the choices of these individuals **in a negative fashion**, meaning that people either do not particularly care about these features or deeply dislike them.

3.7.V CLUSTER 5

	Variable	Method	Variances	t Value	DF	Pr > |t|
1	new_q_1	Satterthwaite	Unequal	-1.38	12.290	0.1926
2	new_q_2	Satterthwaite	Unequal	-1.69	12.490	0.1160
3	new_q_3	Satterthwaite	Unequal	0.52	12.789	0.6146
4	new_q_4	Satterthwaite	Unequal	2.62	13.871	0.0202
5	new_q_5	Satterthwaite	Unequal	0.06	13.373	0.9537
6	new_q_6	Satterthwaite	Unequal	3.98	16.845	0.0010
7	new_q_7	Satterthwaite	Unequal	-1.87	12.675	0.0854
8	new_q_8	Satterthwaite	Unequal	-2.15	12.370	0.0520
9	new_q_9	Satterthwaite	Unequal	2.09	13.063	0.0572
10	new_q_10	Satterthwaite	Unequal	2.69	12.768	0.0188
11	new_q_11	Satterthwaite	Unequal	2.99	13.304	0.0103
12	new_q_12	Satterthwaite	Unequal	-2.22	13.570	0.0439
13	new_q_13	Satterthwaite	Unequal	-4.46	15.626	0.0004
14	new_q_14	Satterthwaite	Unequal	-1.51	13.315	0.1556
15	new_q_15	Satterthwaite	Unequal	0.93	15.236	0.3678
16	new_q_16	Satterthwaite	Unequal	2.64	12.607	0.0208
17	new_q_17	Satterthwaite	Unequal	2.12	12.434	0.0550
18	new_q_18	Satterthwaite	Unequal	-0.44	14.780	0.6679
19	new_q_19	Satterthwaite	Unequal	0.12	16.910	0.9096
20	new_q_20	Satterthwaite	Unequal	0.48	14.237	0.6365
21	new_q_21	Satterthwaite	Unequal	1.71	14.294	0.1091

Noteworthy variables here are the possibility of having various types of *Plans* (new_q_6), which **positively** impacts the behaviour of this cluster, and *Saving money* (new_q_13), which also does but in a **negative way**.

CHI-SQUARE TEST FOR SOCIODEMOGRAPHIC VARIABLES

In order to better understand clusters, we want to know if there is a relationship between them and the sociodemographic variables.
We apply the chi-square approach which is the method used for 2 categorical variables relation.
We want to know if each cluster and the sociodemographic variables are independent, which happens when the expected and the observed distributions are equal.
The chi-square measures the squared distance of data. As with the t-test, to each chi-square, a p-value corresponds, which is the probability that H0, the situation of independence, holds. The null hypothesis, the hypothesis that the partition in clusters is independent of the other variable and the chi-square is not 0 due to a random effect, can be accepted if the p-value is higher than 0.05. SAS has calculated chi-square with the following command:

```
proc freq data=streaming_cluster;
 table (d_1 d_2 d_3 d_4 d_5 d_6 d_7 d_8 d_9 )*cluster/expected chisq;
 run;
```

Among our sociodemographic variables (age, gender, status, family members, place of birth, place of residence, income, level of education, and the number of minors in the family) only **minors in the family can be** important in the partitioning of our clusters since its p-value is lower than 0.05.
As we can see in the table below, in cluster 1 we have an overrepresentation of families with no minors since the observed value is higher than the expected value, while in cluster 3 we have an underrepresentation of families with minors.

Frequency
Expected
Percent
Row Pct
Col Pct

Table of d_9 by CLUSTER

d_9(d_9)	CLUSTER 1	2	3	4	5	Total
No	39 33.824 38.24 52.00 84.78	9 14.706 8.82 12.00 45.00	8 7.3529 7.84 10.67 80.00	11 10.294 10.78 14.67 78.57	8 8.8235 7.84 10.67 66.67	75 73.53
Yes	7 12.176 6.86 25.93 15.22	11 5.2941 10.78 40.74 55.00	2 2.6471 1.96 7.41 20.00	3 3.7059 2.94 11.11 21.43	4 3.1765 3.92 14.81 33.33	27 26.47
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

Frequency Missing = 9

Statistics for Table of d_9 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	12.0447	0.0170

BEHAVIOURAL VARIABLES ANALYSIS

To incorporate behavioral variables in the analysis of the clusters we could not use binary correspondence analysis, which is a factor transformation that transforms a frequency table into a continuous space generating coordinates, due to the lack of a distribution of expenditures to compare the total distribution to (using rarity and chi-square). In fact, in order to study the distance between an item and total expenditure, to use this factor we should transform data eliminating absolute values and taking into consideration only relative values since each item has some degree of importance generated by rarity.

since we could not use the binary correspondence method, we decided to use a different one and simply merged the data of our previously created clusters (found in the PCA) with our behavioral variables (*):

```
data behavioral_cluster;
MERGE clus streaming;
by id;
run;
proc means data=behavioral_cluster;
var b_1-b_9;
class cluster;
run;
```

And then we computed the chi-square:

```
proc freq data=behavioral_cluster;
 table (b_1 b_2 b_3 b_4 b_5 b_6 b_7 b_8 b_9)*cluster/expected chisq;
 run;
```

(*) We have also done this during the analysis of the sociodemographic variables.

Based on which variable has a lower p-value than 0.05, we found that the behavioral ones characterizing our clusters are:

Type of streaming service used (b_1):

The FREQ Procedure

Table of b_1 by CLUSTER

Frequency Expected Percent Row Pct Col Pct	CLUSTER					
b_1(b_1)	1	2	3	4	5	Total
Amazon Prime Video, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Amazon Prime Video, NowTV, Illegal streaming platforms	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Anime website	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Hulu	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
Illegal streaming platforms	1 2.2549 0.98 20.00 2.17	0 0.9804 0.00 0.00 0.00	3 0.4902 2.94 60.00 30.00	1 0.6863 0.98 20.00 7.14	0 0.5882 0.00 0.00 0.00	5 4.90
Illegal streaming platforms, Dowload by Torrent	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Infinity	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix	2 7.6667 1.96 11.76 4.35	10 3.3333 9.80 58.82 50.00	0 1.6667 0.00 0.00 0.00	0 2.3333 0.00 0.00 0.00	5 2 4.90 29.41 41.67	17 16.67
Netflix, Amazon Prime Video	6 4.5098 5.88 60.00 13.04	1 1.9608 0.98 10.00 5.00	0 0.9804 0.00 0.00 0.00	1 1.3725 0.98 10.00 7.14	2 1.1765 1.96 20.00 16.67	10 9.80
Netflix, Amazon Prime Video, Disney +	8 8.5686 7.84 42.11 17.39	4 3.7255 3.92 21.05 20.00	1 1.8627 0.98 5.26 10.00	4 2.6078 3.92 21.05 28.57	2 2.2353 1.96 10.53 16.67	19 18.63
Netflix, Amazon Prime Video, Disney +, Crunchy roll	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, DAZN	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, DAZN, RaiPlay, sky go, Mediaset infinity	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Dazn, Infinity+	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, HBO Max	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, HBO Max, Hulu, NowTV, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, HBO Max, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Hulu	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Illegal streaming platforms	9 4.5098 8.82 90.00 19.57	0 1.9608 0.00 0.00 0.00	1 0.9804 0.98 10.00 10.00	0 1.3725 0.00 0.00 0.00	0 1.1765 0.00 0.00 0.00	10 9.80
Netflix, Amazon Prime Video, Disney +, Illegal streaming platforms, Twitch	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, NowTV	2 1.3529 1.96 66.67 4.35	0 0.5882 0.00 0.00 0.00	0 0.2941 0.00 0.00 0.00	0 0.4118 0.00 0.00 0.00	1 0.3529 0.98 33.33 8.33	3 2.94
Netflix, Amazon Prime Video, Disney +, NowTV, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Paramount plus	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, TimVision	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Hulu, U-NEXT	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Illegal streaming platforms	2 1.3529 1.96 66.67 4.35	0 0.5882 0.00 0.00 0.00	0 0.2941 0.00 0.00 0.00	0 0.4118 0.00 0.00 0.00	1 0.3529 0.98 33.33 8.33	3 2.94
Netflix, Amazon Prime Video, Illegal streaming platforms, SkyGo	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Now TV	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, mubi	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Disney +	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Illegal streaming platforms	5 2.7059 4.90 83.33 10.87	0 1.1765 0.00 0.00 0.00	0 0.5882 0.00 0.00 0.00	0 0.8235 0.00 0.00 0.00	1 0.7059 0.98 16.67 8.33	6 5.88
Netflix, Illegal streaming platforms, * I used Netflix before, but now it doesn't work in my country, so I use just some streaming services of my country??	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Illegal streaming platforms, Mubi	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Prime video	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Žmonės cinema, various cinema festival sites (Ex. Kino pavasaris, NKdoku)	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
None	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Sky Go	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Telegram	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

Statistic	DF	Value	Prob
Chi-Square	144	194.5224	0.0032

Hours per week spent on legal streaming platforms (b_2):

Frequency Expected Percent Row Pct Col Pct	Table of b_2 by CLUSTER					
	CLUSTER					
b_2(b_2)	1	2	3	4	5	Total
1-4 hours	13 18.49 12.75 31.71 28.26	5 8.0392 4.90 12.20 25.00	4 4.0196 3.92 9.76 40.00	11 5.6275 10.78 26.83 78.57	8 4.8235 7.84 19.51 66.67	41 40.20
30 mins	3 3.1569 2.94 42.86 6.52	2 1.3725 1.96 28.57 10.00	2 0.6863 1.96 28.57 20.00	0 0.9608 0.00 0.00 0.00	0 0.8235 0.00 0.00 0.00	7 6.86
5+ hours	27 20.745 26.47 58.70 58.70	12 9.0196 11.76 26.09 60.00	1 4.5098 0.98 2.17 10.00	2 6.3137 1.96 4.35 14.29	4 5.4118 3.92 8.70 33.33	46 45.10
I don't use streaming services	3 3.6078 2.94 37.50 6.52	1 1.5686 0.98 12.50 5.00	3 0.7843 2.94 37.50 30.00	1 1.098 0.98 12.50 7.14	0 0.9412 0.00 0.00 0.00	8 7.84
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
Frequency Missing = 9						

Statistics for Table of b_2 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	12	31.0297	0.0019

The option *1-4 hours* is overrepresented in clusters 4 and 5 and underrepresented in clusters 1 and 2 (we understand that by comparing the frequency and the expected value).
The opposite is true for the option *5+ hours*.
The other two options are more or less what was expected.

***Hours per week spent on Netflix (b_3)*:**

Frequency Expected Percent Row Pct Col Pct	Table of b_3 by CLUSTER					
	CLUSTER					
b_3(b_3)	1	2	3	4	5	Total
1-4 hours	27 23.451 26.47 51.92 58.70	5 10.196 4.90 9.62 25.00	4 5.098 3.92 7.69 40.00	8 7.1373 7.84 15.38 57.14	8 6.1176 7.84 15.38 66.67	52 50.98
30 mins	1 3.1569 0.98 14.29 2.17	3 1.3725 2.94 42.86 15.00	0 0.6863 0.00 0.00 0.00	2 0.9608 1.96 28.57 14.29	1 0.8235 0.98 14.29 8.33	7 6.86
5+ hours	13 13.078 12.75 44.83 28.26	10 5.6863 9.80 34.48 50.00	0 2.8431 0.00 0.00 0.00	3 3.9804 2.94 10.34 21.43	3 3.4118 2.94 10.34 25.00	29 28.43
I don't watch Netflix	5 6.3137 4.90 35.71 10.87	2 2.7451 1.96 14.29 10.00	6 1.3725 5.88 42.86 60.00	1 1.9216 0.98 7.14 7.14	0 1.6471 0.00 0.00 0.00	14 13.73
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
Frequency Missing = 9						

Statistics for Table of b_3 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	12	33.9296	0.0007

As for the *1-4 hours* option, we have an overrepresentation in clusters 1, 4, and 5 and an underrepresentation in clusters 2 and 3.
When it comes to the *30 mins* option, it is overall as expected. The same is true for the last two options but we have a slight overrepresentation of the third option in cluster 2, and of the fourth option in cluster 3.

Hours per week spent on illegal platforms (b_4):

Table of b_4 by CLUSTER

Frequency
Expected
Percent
Row Pct
Col Pct

b_4(b_4)	CLUSTER 1	2	3	4	5	Total
1-4 hours	14 10.373 13.73 60.87 30.43	0 4.5098 0.00 0.00 0.00	5 2.2549 4.90 21.74 50.00	0 3.1569 0.00 0.00 0.00	4 2.7059 3.92 17.39 33.33	23 22.55
30 mins	7 6.3137 6.86 50.00 15.22	2 2.7451 1.96 14.29 10.00	2 1.3725 1.96 14.29 20.00	1 1.9216 0.98 7.14 7.14	2 1.6471 1.96 14.29 16.67	14 13.73
5+ hours	5 4.0588 4.90 55.56 10.87	0 1.7647 0.00 0.00 0.00	2 0.8824 1.96 22.22 20.00	1 1.2353 0.98 11.11 7.14	1 1.0588 0.98 11.11 8.33	9 8.82
I don't use illegal streaming services	20 25.255 19.61 35.71 43.48	18 10.98 17.65 32.14 90.00	1 5.4902 0.98 1.79 10.00	12 7.6863 11.76 21.43 85.71	5 6.5882 4.90 8.93 41.67	56 54.90
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

Frequency Missing = 9

Statistics for Table of b_4 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	12	29.4812	0.0033

For this variable, we can notice an overrepresentation of option *1-4 hours* in the first and fifth clusters. Also, we observe an underrepresentation of the last option in cluster 1, as opposed to an overrepresentation in clusters 2 and 4.

The other two options are very close to what was expected.

Sharing their Netflix account wih someone (b_6):

Table of b_6 by CLUSTER

Frequency
Expected
Percent
Row Pct
Col Pct

b_6(b_6)	CLUSTER 1	2	3	4	5	Total
Family	24 25.706 23.53 42.11 52.17	13 11.176 12.75 22.81 65.00	3 5.5882 2.94 5.26 30.00	7 7.8235 6.86 12.28 50.00	10 6.7059 9.80 17.54 83.33	57 55.88
Friends	17 12.627 16.67 60.71 36.96	3 5.4902 2.94 10.71 15.00	1 2.7451 0.98 3.57 10.00	6 3.8431 5.88 21.43 42.86	1 3.2941 0.98 3.57 8.33	28 27.45
I don't have an account	4 5.8627 3.92 30.77 8.70	2 2.549 1.96 15.38 10.00	6 1.2745 5.88 46.15 60.00	1 1.7843 0.98 7.69 7.14	0 1.5294 0.00 0.00 0.00	13 12.75
Nobody	1 1.8039 0.98 25.00 2.17	2 0.7843 1.96 50.00 10.00	0 0.3922 0.00 0.00 0.00	0 0.549 0.00 0.00 0.00	1 0.4706 0.98 25.00 8.33	4 3.92
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

Frequency Missing = 9

Statistics for Table of b_6 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	12	33.7597	0.0007

Lastly, for this variable, the option *Friends* presents is slightly overrepresented in clusters 1 and 4, the same is true for the option *I don't have an account* but for cluster 3 and the rest is what was expected.

LIFESTYLE VARIABLES AND MULTIPLE CORRESPONDENCE ANALYSIS (MCA)

6.1 MULTIPLE CORRESPONDENCE ANALYSIS

Now we use **lifestyle variables** that are in the form of *YES-NO* answers to describe clusters. Since these variables are categorical, we have a qualitative space.

In order to analyze these binary data, we perform **Multiple Correspondence Analysis**, which is a factor transformation that transforms these binary variables into coordinates of a quantitative continuous space with orthogonal axes.

The metric to measure the distance between coordinates is based on the *rarity* of a phenomenon, which means that two units are more similar if they have common characteristics that are rare. A category must be overvalued if it has few units.

The first thing we did was change our responses in the **binary language** (1;0), where *yes=1 and no=0*.

Then we ran the *proc corresp* procedure to find the coordinates of our units in the quantitative space, choosing a specific number of dimensions.

The number of dimensions is to be chosen with the formula:

'n' dimensions= ('n' categories - 'n' variables)-1

(12-6) -1= 5

However, since 5 dimensions were too many for our analysis in SAS, we used 4.

```
proc corresp data=streaming_lifestyle binary outc=mca_coord dim=4;
 var l_11 l_22 l_33 l_44 l_55 l_66;
 id id;
 run;
```

6.2 INERTIA AND CHI-SQUARE DECOMPOSITION

The CORRESP Procedure

Inertia and Chi-Square Decomposition

Singular Value	Principal Inertia	Chi-Square	Percent	Cumulative Percent
0.35951	0.12925	48.726	41.38	41.38
0.30357	0.09215	34.741	29.50	70.88
0.25175	0.06338	23.894	20.29	91.17
0.16604	0.02757	10.394	8.83	100.00
	0.31235	117.755	100.00	

Degrees of Freedom = 396

The **correspondence analysis** calculates a set of orthogonal dimensions, each of which represents a portion of the **total inertia**, that is, the *total variability*.

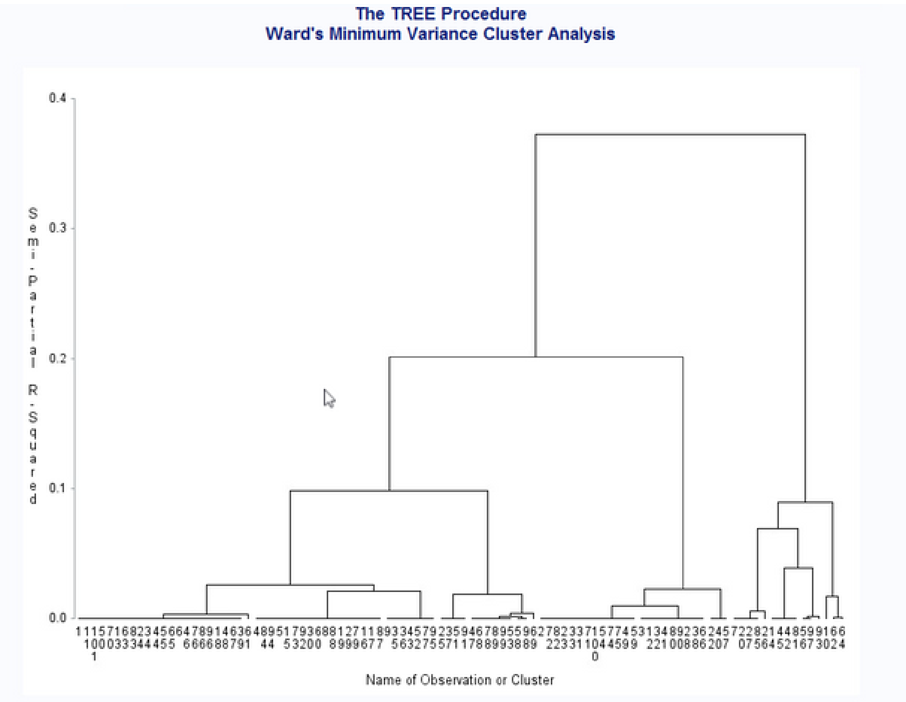
The benchmark, in this case, is calculated as:

total chi-square / 'n' eigenvalues

117.755 / 4 = 29.4

We select just the first two dimensions since their chi-square is higher than the benchmark, which enables us to comprehend 70.88% of the variability present when generating our clusters. We ran the ***proc cluster*** procedure, which convinced us to choose 5 clusters, and the ***proc tree*** procedure to output the data in the 5 clusters.

```
proc cluster data=mca_coord method=ward dim=2 outtree=tree_mca;
 where _TYPE_='OBS';
 id id;
 run;
proc tree data=tree_mca;run;
proc tree data=tree_mca nclusters=5 out=cluster_mca noprint;
 id id;
 run;
```



Next, after changing id into a numerical variable and having sorted the two datasets by id, we merged the 5 clusters with the lifestyle dataset to go on with the description of the content of the clusters.

```
proc sort data=clus;by id;run;
proc sort data=lifestyle2;by id;run;

data streaming_lifestyle;
MERGE clus lifestyle2;
by id;
run;
```

6.3 CHI-SQUARE TEST

To analyze the relationship between our partition in clusters and the lifestyle variables we generated **cross-tables** (one for each variable) with the **proc freq** command.
In these tables, we find the observed distribution and the expected distribution according to the independence situation for each variable among the clusters.

```
proc freq data=streaming_lifestyle;
 table (l_11 l_22 l_33 l_44 l_55 l_66)*cluster /expected chisq;
 run;
```

In the tables below we can see that none of our variables, which we transformed into binary data previously, influences the partition in clusters because their p-values are higher than 0.05, which leads us to ***fail to reject the null hypothesis***.

The FREQ Procedure

Table of l_11 by CLUSTER

Frequency / Expected / Percent / Row Pct / Col Pct	l_11(l_11)	CLUSTER 1	2	3	4	5	Total
Frequency	0	11	6	2	1	1	21
Expected		9.4706	4.1176	2.0588	2.8824	2.4706	
Percent		10.78	5.88	1.96	0.98	0.98	20.59
Row Pct		52.38	28.57	9.52	4.76	4.76	
Col Pct		23.91	30.00	20.00	7.14	8.33	
Frequency	1	35	14	8	13	11	81
Expected		36.529	15.882	7.9412	11.118	9.5294	
Percent		34.31	13.73	7.84	12.75	10.78	79.41
Row Pct		43.21	17.28	9.88	16.05	13.58	
Col Pct		76.09	70.00	80.00	92.86	91.67	
Frequency	Total	46	20	10	14	12	102
Percent		45.10	19.61	9.80	13.73	11.76	100.00

Frequency Missing = 9

Statistics for Table of l_11 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	4.0470	0.3997

Frequency
Expected
Percent
Row Pct
Col Pct

Table of I_22 by CLUSTER

I_22(I_22)	CLUSTER 1	2	3	4	5	Total
0	11	4	4	3	0	22
	9.9216	4.3137	2.1569	3.0196	2.5882	
	10.78	3.92	3.92	2.94	0.00	21.57
	50.00	18.18	18.18	13.64	0.00	
	23.91	20.00	40.00	21.43	0.00	
1	35	16	6	11	12	80
	36.078	15.686	7.8431	10.98	9.4118	
	34.31	15.69	5.88	10.78	11.76	78.43
	43.75	20.00	7.50	13.75	15.00	
	76.09	80.00	60.00	78.57	100.00	
Total	46	20	10	14	12	102
	45.10	19.61	9.80	13.73	11.76	100.00

Frequency Missing = 9

Statistics for Table of I_22 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	5.4869	0.2409

Frequency
Expected
Percent
Row Pct
Col Pct

Table of I_33 by CLUSTER

I_33(I_33)	CLUSTER 1	2	3	4	5	Total
0	12	9	4	3	4	32
	14.431	6.2745	3.1373	4.3922	3.7647	
	11.76	8.82	3.92	2.94	3.92	31.37
	37.50	28.13	12.50	9.38	12.50	
	26.09	45.00	40.00	21.43	33.33	
1	34	11	6	11	8	70
	31.569	13.725	6.8627	9.6078	8.2353	
	33.33	10.78	5.88	10.78	7.84	68.63
	48.57	15.71	8.57	15.71	11.43	
	73.91	55.00	60.00	78.57	66.67	
Total	46	20	10	14	12	102
	45.10	19.61	9.80	13.73	11.76	100.00

Frequency Missing = 9

Statistics for Table of I_33 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	3.3321	0.5039

Frequency
Expected
Percent
Row Pct
Col Pct

Table of I_44 by CLUSTER

I_44(I_44)	CLUSTER 1	2	3	4	5	Total
0	7	4	1	1	1	14
	6.3137	2.7451	1.3725	1.9216	1.6471	
	6.86	3.92	0.98	0.98	0.98	13.73
	50.00	28.57	7.14	7.14	7.14	
	15.22	20.00	10.00	7.14	8.33	
1	39	16	9	13	11	88
	39.686	17.255	8.6275	12.078	10.353	
	38.24	15.69	8.82	12.75	10.78	86.27
	44.32	18.18	10.23	14.77	12.50	
	84.78	80.00	90.00	92.86	91.67	
Total	46	20	10	14	12	102
	45.10	19.61	9.80	13.73	11.76	100.00

Frequency Missing = 9

Statistics for Table of I_44 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	1.6755	0.7952

Frequency Expected Percent Row Pct Col Pct	Table of I_55 by CLUSTER					
	CLUSTER					
I_55(I_55)	1	2	3	4	5	Total
0	29 25.706 28.43 50.88 63.04	8 11.176 7.84 14.04 40.00	7 5.5882 6.86 12.28 70.00	9 7.8235 8.82 15.79 64.29	4 6.7059 3.92 7.02 33.33	57 55.88
1	17 20.294 16.67 37.78 36.96	12 8.8235 11.76 26.67 60.00	3 4.4118 2.94 6.67 30.00	5 6.1765 4.90 11.11 35.71	8 5.2941 7.84 17.78 66.67	45 44.12
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
Frequency Missing = 9						

Statistics for Table of I_55 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	6.6874	0.1534

Frequency Expected Percent Row Pct Col Pct	Table of I_66 by CLUSTER					
	CLUSTER					
I_66(I_66)	1	2	3	4	5	Total
0	17 15.784 16.67 48.57 36.96	8 6.8627 7.84 22.86 40.00	2 3.4314 1.96 5.71 20.00	6 4.8039 5.88 17.14 42.86	2 4.1176 1.96 5.71 16.67	35 34.31
1	29 30.216 28.43 43.28 63.04	12 13.137 11.76 17.91 60.00	8 6.5686 7.84 11.94 80.00	8 9.1961 7.84 11.94 57.14	10 7.8824 9.80 14.93 83.33	67 65.69
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
Frequency Missing = 9						

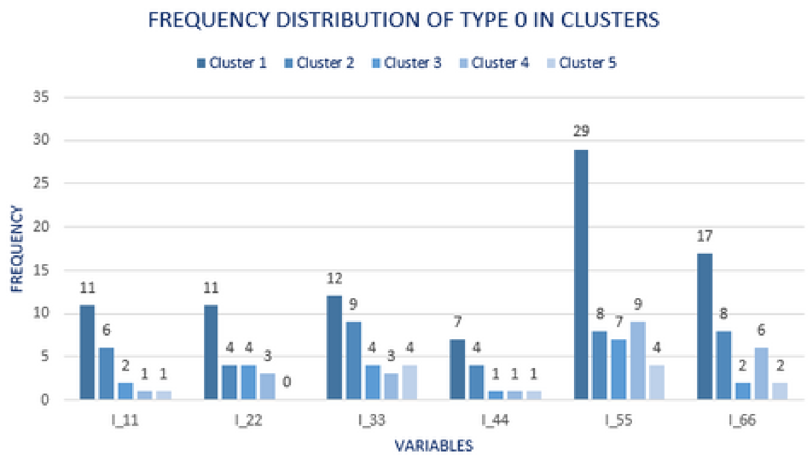
Statistics for Table of I_66 by CLUSTER

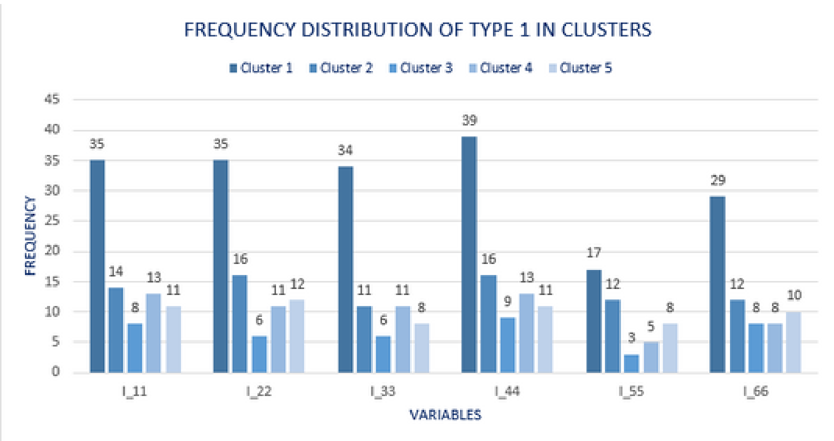
Statistic	DF	Value	Prob
Chi-Square	4	3.4498	0.4856

After that, we used the proc means procedure through which we created a compact table that shows the number of people who answered yes (1) and no (0) for each variable in each cluster. Thus, in the sixth and 12th rows, we see the marginal distribution, and in the other rows the distribution of the variables in the single clusters.

```
proc means data=streaming_lifestyle sum;
var l_11 l_22 l_33 l_44 l_55 l_66;
output out=lifestyle_collapsed sum=;
class cluster;
run;
```

	CLUSTER	_TYPE_	_FREQ_	l_11	l_22	l_33	l_44	l_55	l_66
	1	0		11	11	12	7	29	17
	2	0		6	4	9	4	8	8
	3	0		2	4	4	1	7	2
	4	0		1	3	3	1	9	6
	5	0		1	0	4	1	4	2
TOT				21	22	32	14	57	35
	1	1		35	35	34	39	17	29
	2	1		14	16	11	16	12	12
	3	1		8	6	6	9	3	8
	4	1		13	11	11	13	5	8
	5	1		11	12	8	11	8	10
TOT		1	102	81	80	70	88	45	67

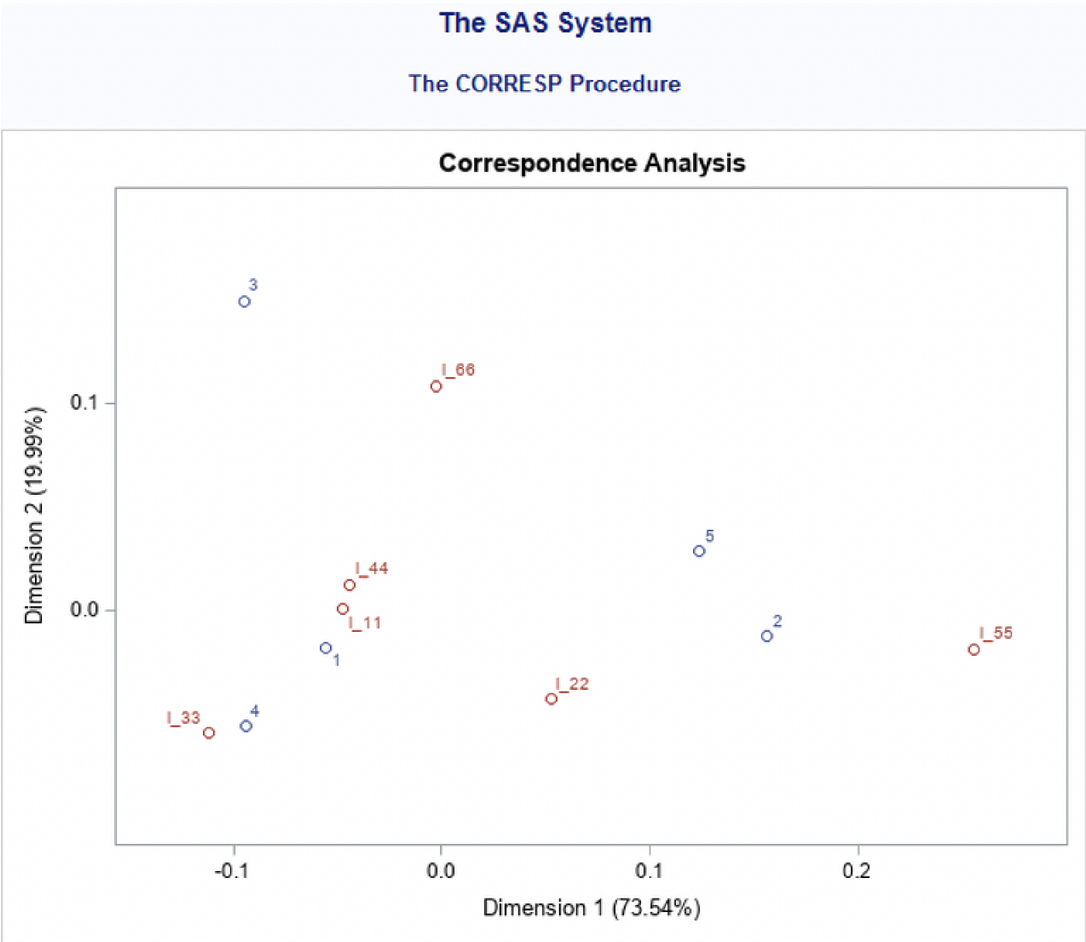




6.4 THE CORRESPONDENCE MAP

Then with the **proc corresp** command, we represented in an orthogonal space the position of our clusters with respect to the lifestyle variables.

```
proc corresp data=lifestyle_collapsed;
 var l_11 l_22 l_33 l_44 l_55 l_66;
 id cluster;
 where _TYPE_=1;
 run;
```



Cluster 3 and 5 are far from the variables. Cluster 1 is characterized by both l_11 and l_44, while cluster 4 is characterized by l_33.

DISCRIMINANT ANALYSIS LIFESTYLE AND THE PREDICTIVE MODEL

Opinion clusterization is the best way to create clusters.
Of course, a considerable problem arises with the **dimensionality of information**, as we studied the opinions only of a small sample of people, not of the entire potential customer base.
Companies also want to predict opinion clusters for the units that haven't been interviewed using other data (such as behavioral and lifestyle data). In fact, big data are helpful to predict small data (opinions).
In order to do so, they must create a **predictive model** that explains the relationship between opinion clusters and the other variables.
This model has the function:

Opinions = f [behaviors, lifestyle, ...]

in which behaviors and lifestyle represent the independent variables, that is to say, the **predictors**. Opinions are instead the dependent variables, meaning the **target variables**.
With the aim of finding the function that maximizes the fitting likelihood of the relationship between these variables, we used a geometrical approach called the **k-nearest neighbor (KNN)** approach, which **assigns units to clusters on the basis of the proximity they have to existing clusters by using a selected number of "k" nearest neighbors**. The k value in the k-NN algorithm defines how many neighbors will be checked to determine the classification of a specific query point.
Defining k can be a **balancing act** (tradeoff between bias error and variance) as different values can lead to overfitting or underfitting. Lower values of k can have high variance, but low bias, and larger values of k may lead to high bias and lower variance as we use more information. The choice of k will largely depend on the input data as data with more outliers or noise will likely perform better with higher values of k. Overall, it is recommended to have an odd number for k to avoid ties in classification, and cross-validation tactics can help you choose the optimal k for your dataset.

We merged the coordinates we obtained from the previously performed MCA with the clusters we obtained from the PCA, which are related to opinions.

```
proc sort data=input_var; by id_num; run;
proc sort data= mca_coord_rows; by id_num; run;
data input_var_1;
merge input_var mca_coord_rows;
by id_num;
run;
```

Afterward, we proceeded with the **discriminant analysis** using the **proc discrim** command. The discriminant analysis is useful to **develop a discriminant criterion on the basis of which units are classified into different groups**. It also calculates **error rates** as the probability that units are misplaced in clusters and therefore evaluates **if the discriminant criterion is effective**. The dataset used to derive the discriminant criterion is the training dataset.

The **cross-validation method** is mainly used in settings where the goal is prediction, and one wants to estimate how accurately a predictive model will perform in practice. It takes advantage of two datasets: a 'training dataset' used to train the model and a 'testing dataset' of unknown data used to validate the model.

We ran a **macro procedure** to perform a discriminant analysis with different values of 'k' as to find the 'k' with the smallest error rate.

To be able to find the interval of 'k' we use the Enas and Choi formula:

Correct interval between k=n^2/8 and k=n^3/8

This formula is for dichotomous situations, so we have to relativize it for our 5 clusters.

								K	
N	111		0,25	=1/4	3,245867	=C3^E3	3	7,5	=I3*C7
			0,375	=3/8	5,847849	=C3^E4	6	15	=I4*C7
clusters	5								
corr factor	2,5	clusters/2							

```
%MACRO DO_kk;
%do k=8 %to 15;
proc discrim data=input_var_1 crossvalidate
            method=NPAR k=&k outcross=unico_testout_&k posterr
            ;
      class cluster;
      var dim:;
      ods output ErrorCrossVal=matrix_&k;
    run;
    data matrix_&k; set matrix_&k; kappa=&k; run;
%end;
%mend do_kk;
%do_kk;
```

```
data matrixone(where=(type='Rate'));
set matrix_8-matrix_15;
run;
```

In our case, the 'k' with the lowest error rate (0.6895) is k=12.

	Type	1	2	3	4	5	Total	kappa
1	Rate	0.8261	0.5000	0.3000	1.0000	0.9167	0.7086	8
2	Rate	0.9348	0.6500	0.3000	1.0000	0.9167	0.7603	9
3	Rate	1.0000	0.8000	0.3000	1.0000	0.8333	0.7867	10
4	Rate	1.0000	0.8000	0.3000	0.7857	0.7500	0.7271	11
5	Rate	0.9783	0.8000	0.3000	0.7857	0.5833	0.6895	12
6	Rate	0.8696	0.8000	0.6000	0.7857	0.5000	0.7111	13
7	Rate	0.8696	0.8000	0.6000	1.0000	0.6667	0.7872	14
8	Rate	0.8478	0.8000	0.6000	0.7857	0.6667	0.7400	15

Then we ran again the **proc discrim** using k=12 and we obtained a **confusion matrix**.

```
proc discrim data=input_var_1 crossvalidate method=npar k=12 out= prob_result;
var dim:;
class cluster;
run;
```

The DISCRIM Procedure
Classification Summary for Calibration Data: WORK.INPUT_VAR_1
Cross-validation Summary using 12 Nearest Neighbors

Number of Observations and Percent Classified into CLUSTER						
From CLUSTER	1	2	3	4	5	Total
.	1 12.50	1 12.50	3 37.50	0 0.00	3 37.50	8 100.00
1	1 2.17	5 10.87	14 30.43	6 13.04	20 43.48	46 100.00
2	0 0.00	4 20.00	4 20.00	2 10.00	10 50.00	20 100.00
3	1 10.00	1 10.00	7 70.00	1 10.00	0 0.00	10 100.00
4	2 14.29	2 14.29	5 35.71	3 21.43	2 14.29	14 100.00
5	2 16.67	3 25.00	0 0.00	2 16.67	5 41.67	12 100.00
Total	7 6.36	16 14.55	33 30.00	14 12.73	40 36.36	110 100.00
Priors	0.2	0.2	0.2	0.2	0.2	

Error Count Estimates for CLUSTER						
	1	2	3	4	5	Total
Rate	0.9783	0.8000	0.3000	0.7857	0.5833	0.6895
Priors	0.2000	0.2000	0.2000	0.2000	0.2000	

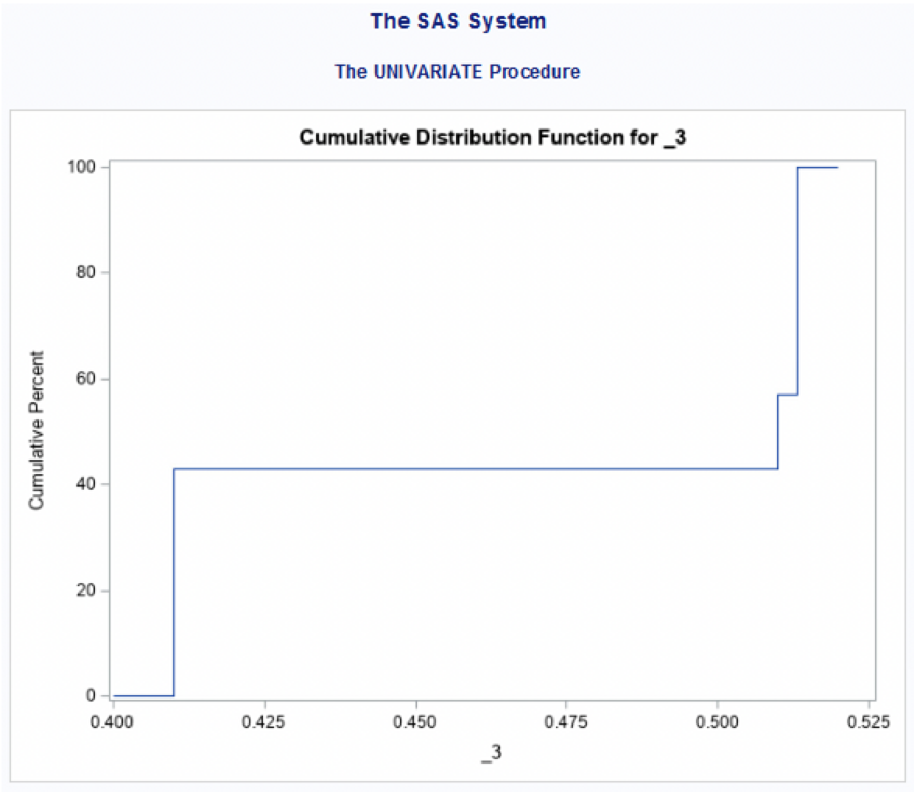
The row from the table represents the destination cluster and the column represents the original cluster.
In the main diagonal, we have the probability that units belonging to a cluster will be correctly predicted in that cluster.
In the error count table, we have the error rate for each cluster.
We see that the possibility that a unit belonging to cluster 1, 2, and 4 is misplaced is actually very high, with error rates of 97.83%, 80.00%, and 78.57% respectively.
Meantime, the error is lower for cluster 5 and it's the lowest for cluster 3 (30%), thus the probability that the model predicts units in that cluster correctly is quite high.

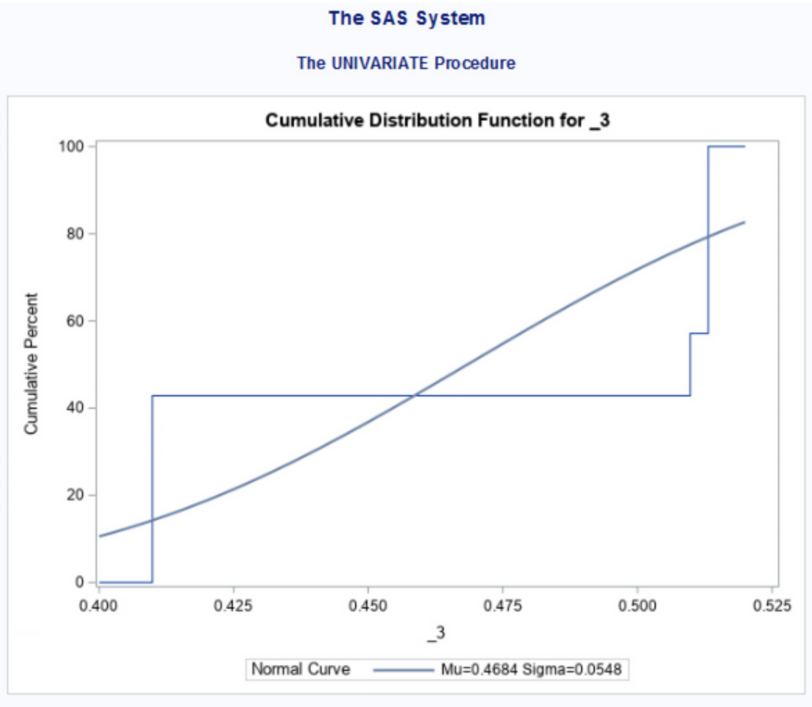
Now we compute the **lift**, which is **the ratio of the probability of clustering units correctly with knowledge about lifestyles to the probability without knowledge** (prior probability). It shows us the gain of assigning units to clusters using our model, instead of just guessing.

LIFT 1 = 0.0217/0.20 = 0.1085
LIFT 2 = 0.20/0.20 = 1.0000
LIFT 3 = 0.70/0.20 = 3.5000
LIFT 4 = 0.1243/0.20 = 0.6215
LIFT 5 = 0.4167/0.20 = 2.0835

With the **proc univariate** procedure, we obtained cumulative distribution functions for clusters 3 and 5, which were the clusters that proved to have the highest lifts.

```
proc univariate data= prob_result;
var _3;
where cluster=3 and _into_=3;
cdfplot; run;
```

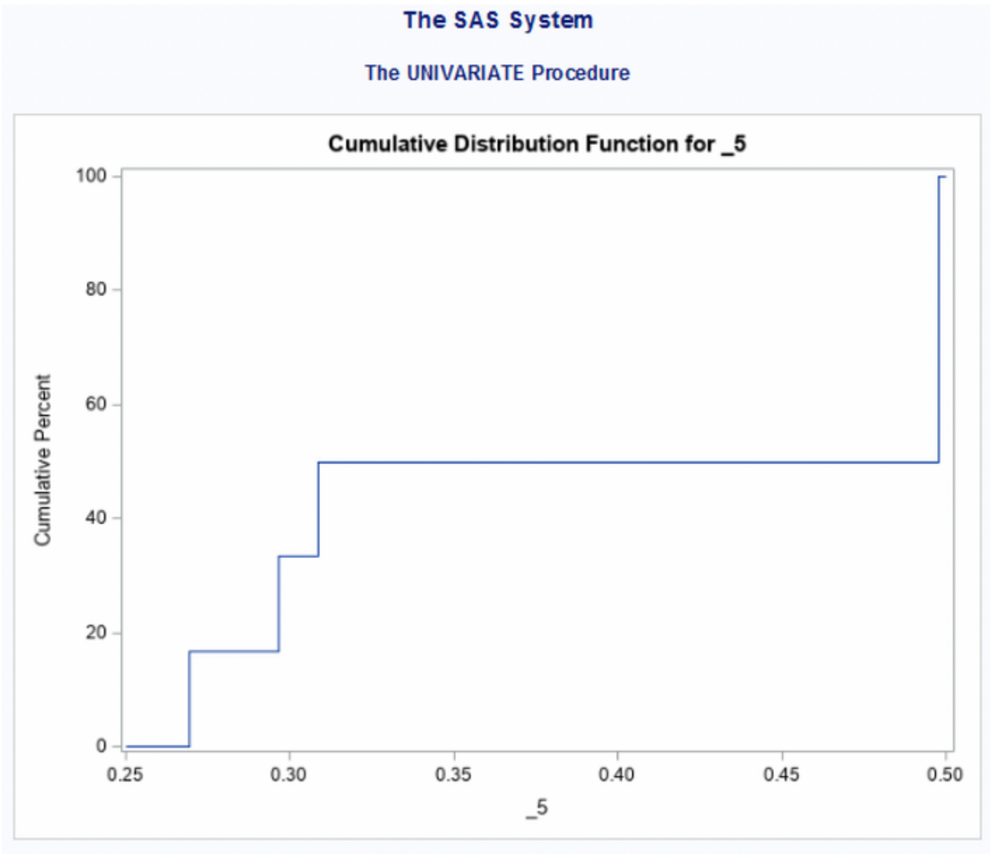


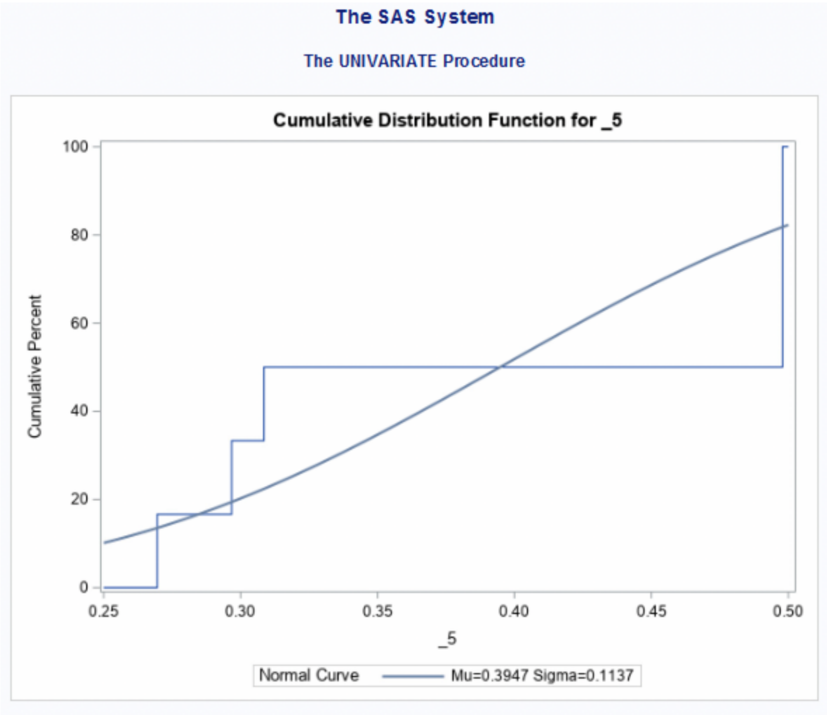


Quantiles (Definition 5)	
Level	**Quantile**
100% Max	0.513147
99%	0.513147
95%	0.513147
90%	0.513147
75% Q3	0.513147
50% Median	0.509830
25% Q1	0.409901
10%	0.409901
5%	0.409901
1%	0.409901
0% Min	0.409901

The cumulative distribution function for cluster 3 shows that the posterior probability of 80% (when there's gain of knowledge) of the units is close to 0.51.

```
proc univariate data= prob_result;
var _5;
where cluster=5 and _into_=5;
cdfplot; run;
```





Quantiles (Definition 5)	
Level	Quantile
100% Max	0.497835
99%	0.497835
95%	0.497835
90%	0.497835
75% Q3	0.497835
50% Median	0.403162
25% Q1	0.296719
10%	0.269591
5%	0.269591
1%	0.269591
0% Min	0.269591

The cumulative distribution function for cluster 5 shows that the posterior probability of 80% (when there's gain of knowledge) of the units is close to 0.50.

CLUSTER DESCRIPTION

8.1 CLUSTER 1 : PIRATE ITALIAN STUDENTS

Cluster 1 is composed of 46 people out of the 111 respondents to our survey, Mostly aged between **18-25.** More than half (56%) of them are studying or working females. The highest level of education for the majority was High School or Bachelor's Degree. The majority of them are residents and citizens of **Italy**. More than half (53%) of their monthly income was **less than 1000 EUR** or between **1000-2000 EUR.** Their families are mainly composed of 2 or 4 family members and most of them (85%) **do not have minors** in their families.
The cluster primarily uses legal streaming platforms like Netflix and Amazon prime. They tend to spend more than **5 hours a week** watching legal streaming platforms, and **1-4 hours watching Netflix**. Less than half of them (**43% of them**) **do not use illegal streaming** platforms. The ones using illegal streaming platforms, prefer it due to the better choices compared to legal streaming platforms. Netflix users share their accounts mostly with their **family members** (52%) and **friends** (37%). Both: legal and illegal streaming platform users watch more (82%) TV series than movies. The cluster common genres are action, adventure, comedy, dramas, and documentaries. The majority of them (76%) characterize themselves as sensitive to price. Most of them (76%) like to stay indoors. 74% of them prefer a healthy lifestyle and 84% consider leisure time important. 63% of them does not characterize themselves as risk-takers and 63% feel stressed in their daily life.

8.2 CLUSTER 2: OLDER SISTERS ARE PLATFORMS LOVERS

Cluster 2 consists of 20 people. Mostly **female students** mostly (80%) aged between **18-25**. The highest level of education for the majority was High School or Bachelor's Degree. Most of them are residents and citizens of Italy and Lithuania. Their monthly income is between 1000 and 2000 EUR. Their families are mostly composed of 4 family members and more than half of them have minors in their families.

The cluster mainly uses legal streaming platforms, at most **Netflix (58%).** They tend to spend **more than 5 hours a week** watching Netflix or other legal streaming platforms. Majority of them 90% of them **do not use illegal streaming** platforms. The ones using illegal streaming platforms, prefer it due to the high prices of legal streaming platforms. Netflix users share their accounts mostly with their family members (65%). They watch more (65%) TV series than movies. The cluster common genres are **romance and drama**. The majority of them (70%) characterize themselves as sensitive to **price.** Most of them (80%) like to stay **indoors. More than half** of them prefer a **healthy lifestyle** and **80%** consider leisure time **important. 60% of them characterize** themselves as **risk-takers** and 60% feel **stressed** in their daily life.

8.3 CLUSTER 3: ANIMATION LOVERS BREAK THE RULES

Cluster 3 consists of 10 people, mainly **males** or **non-binary** workers and students, the majority (70%) aged between **18-25** and **30%** between **26-35.** They have gained a Bachelor's Degree or High School diploma. Most of them are residents and citizens of **Lithuania, Belgium, and France.** Most of their monthly income is **less** than **1000** or between **1001** and **2000** EUR. Their families are mostly composed of 3-4 or 1 family member and the majority of them do **not have** minors in their families.

The cluster mainly uses streaming platforms between **1-4 hours a week**. Only 10% of them **do not use illegal streaming** platforms. **60%** of them do not have a **Netflix** account. The ones using illegal streaming platforms, prefer it due to the **better choices** compared to Netflix, Disney+, or other legal platforms. For them, legal platforms are too expensive, they like to randomly watch content and do not feel the need to have a subscription. They watch more (70%) TV-series than movies. The cluster common genres are **animation, comedy, and drama**. The majority of them (80%) characterize themselves as sensitive to **prices**. Most of them (60%) like to stay **indoors. The majority** of them prefer a **healthy lifestyle** and consider leisure time **important. 70% of them** do not characterize themselves as **risk-takers** and 80% feel **stressed** in their daily life.

8.4 CLUSTER 4: HONOR STUDENTS

Cluster 4 consists of 14 people, mostly (57%) aged between **18-25**, **evenly** divided between working or studying (also around 20% doing both) **males** and **females,** and most of them have completed a Bachelor's Degree or High School Diploma**.** Most of them are residents and citizens of **Italy.** 29% of their monthly income is more than 4001 EUR, 21% between 1000-2000 EUR, and less than 1000 EUR. Their families are mostly composed of **4** members and the majority of them do **not have** minors in their families.

The cluster mainly uses streaming platforms between **1-4 hours a week**. Majority (85%) of them **do not use illegal streaming** platforms. The rest of them, using illegal streaming platforms, prefer it due to the **better choices** compared to Netflix, Disney+, or other legal platforms. The majority of them share their Netflix account with their **friends** and **family**. They are **evenly divided between movie and TV series** preferences. The cluster common genres are **action and comedy**. The majority of them (93%) characterize themselves as sensitive to **prices**. Most of them (79%) like to stay **indoors. The majority** of them prefer a **healthy lifestyle** and consider leisure time **important.** More than half (64%) disagree with characterizing themselves as **risk-takers** and more than half (57%) feel **stressed** in their daily life.

8.5 CLUSTER 5: HOMEBODIES

Cluster 5 consists of 12 people, mostly aged between **18-25**, most of whom are working or studying **females** and completed their Bachelor's Degrees. Most of them are citizens of **Italy** and **Lithuania**, respectively living there. Half (50%) of their monthly income is between **1000-2000** EUR. Their families are mostly composed of **3 or 4** members and the majority of them do **not have** minors in their families.

The cluster mainly uses streaming platforms between **1-4 hours a week**. 41% of them do not use illegal streaming platforms. The rest of them, using illegal streaming platforms, prefer it due to the **better choices** and **high prices** of legal streaming platforms. Both legal and illegal streaming platforms users prefer TV series (75%) to movies. The cluster common genres are **documentaries**, **adventures**, **crime,** and **horror**. The majority of them (92%) characterize them as sensitive to **prices**. All of them (100%) like to stay **indoors. The majority** of them prefer a **healthy lifestyle** and consider leisure time **important.** More than half (66%) can name themselves **risk-takers** and almost all of them (10 out of 12) feel **stressed** in their daily life.

CONCLUSION

In conclusion, SAS software was helpful in this project for analyzing and differentiating streaming service users in Europe.
SAS gave us the ability to gather data on respondents' backgrounds, behavior, and lifestyle into 5 main representative clusters of users.

The predictive model doesn't reliably predict the opinions of users in Cluster 1, Cluster 2, and Cluster 4, due to the inputs we provided in the survey. Our questionnaire presents, therefore, itself with some bias: we mainly got young respondents from Italy and Lithuania because of budget and time restrictions. In order to remove or reduce these biases we would need to change the structure of the questions of the survey and the way we shared it with respondents, which was mainly through people we knew and social media.

Although we were able to find 5 clusters, by minimizing internal variability and maximizing external variability among clusters, the clusters we obtained were quite similar in terms of composition, preferences, and attitudes, again due to the fact that the sample of people we distributed our survey to is a convenient one.
To have a realistic representation of reality and an accurate predictive model we would have needed far more respondents and a different distribution channel.

APPENDIX

Graphs for cluster description

- AGE

Table of d_1 by CLUSTER

Frequency Expected Percent Row Pct Col Pct	CLUSTER					
d_1(d_1)	1	2	3	4	5	Total
+50	1 2.2549 0.98 20.00 2.17	2 0.9804 1.96 40.00 10.00	0 0.4902 0.00 0.00 0.00	1 0.6863 0.98 20.00 7.14	1 0.5882 0.98 20.00 8.33	5 4.90
18-25	33 31.118 32.35 47.83 71.74	16 13.529 15.69 23.19 80.00	7 6.7647 6.86 10.14 70.00	8 9.4706 7.84 11.59 57.14	5 8.1176 4.90 7.25 41.67	69 67.65
26-35	9 9.0196 8.82 45.00 19.57	0 3.9216 0.00 0.00 0.00	3 1.9608 2.94 15.00 30.00	4 2.7451 3.92 20.00 28.57	4 2.3529 3.92 20.00 33.33	20 19.61
36-50	2 2.2549 1.96 40.00 4.35	1 0.9804 0.98 20.00 5.00	0 0.4902 0.00 0.00 0.00	1 0.6863 0.98 20.00 7.14	1 0.5882 0.98 20.00 8.33	5 4.90
<18	1 1.3529 0.98 33.33 2.17	1 0.5882 0.98 33.33 5.00	0 0.2941 0.00 0.00 0.00	0 0.4118 0.00 0.00 0.00	1 0.3529 0.98 33.33 8.33	3 2.94
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- GENDER

Table of d_2 by CLUSTER

Frequency Expected Percent Row Pct Col Pct	CLUSTER					
d_2(d_2)	1	2	3	4	5	Total
Female	27 26.157 26.47 46.55 58.70	13 11.373 12.75 22.41 65.00	2 5.6863 1.96 3.45 20.00	7 7.9608 6.86 12.07 50.00	9 6.8235 8.82 15.52 75.00	58 56.86
Gender fluid	2 0.902 1.96 100.00 4.35	0 0.3922 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.2745 0.00 0.00 0.00	0 0.2353 0.00 0.00 0.00	2 1.96
I prefer not to say	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Male	14 16.235 13.73 38.89 30.43	6 7.0588 5.88 16.67 30.00	6 3.5294 5.88 16.67 60.00	7 4.9412 6.86 19.44 50.00	3 4.2353 2.94 8.33 25.00	36 35.29
Non-binary	3 2.2549 2.94 60.00 6.52	0 0.9804 0.00 0.00 0.00	2 0.4902 1.96 40.00 20.00	0 0.6863 0.00 0.00 0.00	0 0.5882 0.00 0.00 0.00	5 4.90
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- STATUS

Table of d_3 by CLUSTER						
	CLUSTER					
d_3(d_3)	1	2	3	4	5	Total
Retiree	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Student	21 20.294 20.59 46.67 45.65	12 8.8235 11.76 26.67 60.00	2 4.4118 1.96 4.44 20.00	5 6.1765 4.90 11.11 35.71	5 5.2941 4.90 11.11 41.67	45 44.12
Student and worker	7 6.7647 6.86 46.67 15.22	1 2.9412 0.98 6.67 5.00	2 1.4706 1.96 13.33 20.00	3 2.0588 2.94 20.00 21.43	2 1.7647 1.96 13.33 16.67	15 14.71
Unemployed	1 1.3529 0.98 33.33 2.17	1 0.5882 0.98 33.33 5.00	1 0.2941 0.98 33.33 10.00	0 0.4118 0.00 0.00 0.00	0 0.3529 0.00 0.00 0.00	3 2.94
Worker	17 17.137 16.67 44.74 36.96	5 7.451 4.90 13.16 25.00	5 3.7255 4.90 13.16 50.00	6 5.2157 5.88 15.79 42.86	5 4.4706 4.90 13.16 41.67	38 37.25
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- EDUCATION

Frequency Expected Percent Row Pct Col Pct

Table of d_4 by CLUSTER						
	CLUSTER					
d_4(d_4)	1	2	3	4	5	Total
Bachelor's Degree	18 19.392 17.65 41.86 39.13	8 8.4314 7.84 18.60 40.00	4 4.2157 3.92 9.30 40.00	6 5.902 5.88 13.95 42.86	7 5.0588 6.86 16.28 58.33	43 42.16
Elementary school	0 0.902 0.00 0.00 0.00	1 0.3922 0.98 50.00 5.00	0 0.1961 0.00 0.00 0.00	0 0.2745 0.00 0.00 0.00	1 0.2353 0.98 50.00 8.33	2 1.96
High School	20 19.392 19.61 46.51 43.48	10 8.4314 9.80 23.26 50.00	4 4.2157 3.92 9.30 40.00	5 5.902 4.90 11.63 35.71	4 5.0588 3.92 9.30 33.33	43 42.16
Master's Degree	8 5.8627 7.84 61.54 17.39	1 2.549 0.98 7.69 5.00	2 1.2745 1.96 15.38 20.00	2 1.7843 1.96 15.38 14.29	0 1.5294 0.00 0.00 0.00	13 12.75
Ph.D. or higher	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Trade School	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- COUNTRY OF RESIDENCE

Frequency
Expected
Percent
Row Pct
Col Pct

Table of d_5 by CLUSTER

d_5(d_5)	CLUSTER 1	2	3	4	5	Total
Afghanistan	0	0	0	0	0	0
	0	0	0	0	0	
	0.00	0.00	0.00	0.00	0.00	0.00
	.	.	.	.	.	
	0.00	0.00	0.00	0.00	0.00	
Australia	1	0	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.98	0.00	0.00	0.00	0.00	0.98
	100.00	0.00	0.00	0.00	0.00	
	2.17	0.00	0.00	0.00	0.00	
Belgium	1	1	1	0	0	3
	1.3529	0.5882	0.2941	0.4118	0.3529	
	0.98	0.98	0.98	0.00	0.00	2.94
	33.33	33.33	33.33	0.00	0.00	
	2.17	5.00	10.00	0.00	0.00	
Canada	0	0	0	0	0	0
	0	0	0	0	0	
	0.00	0.00	0.00	0.00	0.00	0.00
	.	.	.	.	.	
	0.00	0.00	0.00	0.00	0.00	
Denmark	0	1	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.98	0.00	0.00	0.00	0.98
	0.00	100.00	0.00	0.00	0.00	
	0.00	5.00	0.00	0.00	0.00	
France	0	0	0	0	1	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.00	0.00	0.00	0.98	0.98
	0.00	0.00	0.00	0.00	100.00	
	0.00	0.00	0.00	0.00	8.33	
Germany	1	0	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.98	0.00	0.00	0.00	0.00	0.98
	100.00	0.00	0.00	0.00	0.00	
	2.17	0.00	0.00	0.00	0.00	
Greece	0	0	0	1	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.00	0.00	0.98	0.00	0.98
	0.00	0.00	0.00	100.00	0.00	
	0.00	0.00	0.00	7.14	0.00	
Italy	37	10	8	12	4	71
	32.02	13.922	6.9608	9.7451	8.3529	
	36.27	9.80	7.84	11.76	3.92	69.61
	52.11	14.08	11.27	16.90	5.63	
	80.43	50.00	80.00	85.71	33.33	
Lithuania	3	7	1	1	4	16
	7.2157	3.1373	1.5686	2.1961	1.8824	
	2.94	6.86	0.98	0.98	3.92	15.69
	18.75	43.75	6.25	6.25	25.00	
	6.52	35.00	10.00	7.14	33.33	
Russia	1	0	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.98	0.00	0.00	0.00	0.00	0.98
	100.00	0.00	0.00	0.00	0.00	
	2.17	0.00	0.00	0.00	0.00	
United Kingdom	1	1	0	0	3	5
	2.2549	0.9804	0.4902	0.6863	0.5882	
	0.98	0.98	0.00	0.00	2.94	4.90
	20.00	20.00	0.00	0.00	60.00	
	2.17	5.00	0.00	0.00	25.00	
United States	1	0	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.98	0.00	0.00	0.00	0.00	0.98
	100.00	0.00	0.00	0.00	0.00	
	2.17	0.00	0.00	0.00	0.00	
Total	46	20	10	14	12	102
	45.10	19.61	9.80	13.73	11.76	100.00

- COUNTRY OF ORIGIN

d_6(d_6)	CLUSTER 1	2	3	4	5	Total
Afghanistan	0	0	0	0	0	0
	0	0	0	0	0	
	0.00	0.00	0.00	0.00	0.00	0.00
	.	.	.	.	.	
	0.00	0.00	0.00	0.00	0.00	
Albania	1	0	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.98	0.00	0.00	0.00	0.00	0.98
	100.00	0.00	0.00	0.00	0.00	
	2.17	0.00	0.00	0.00	0.00	
Belgium	0	1	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.98	0.00	0.00	0.00	0.98
	0.00	100.00	0.00	0.00	0.00	
	0.00	5.00	0.00	0.00	0.00	
Bosnia and Herzegovina	1	0	0	1	0	2
	0.902	0.3922	0.1961	0.2745	0.2353	
	0.98	0.00	0.00	0.98	0.00	1.96
	50.00	0.00	0.00	50.00	0.00	
	2.17	0.00	0.00	7.14	0.00	
Brazil	0	0	0	0	0	0
	0	0	0	0	0	
	0.00	0.00	0.00	0.00	0.00	0.00
	.	.	.	.	.	
	0.00	0.00	0.00	0.00	0.00	
Estonia	0	1	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.98	0.00	0.00	0.00	0.98
	0.00	100.00	0.00	0.00	0.00	
	0.00	5.00	0.00	0.00	0.00	
France	0	0	1	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.00	0.98	0.00	0.00	0.98
	0.00	0.00	100.00	0.00	0.00	
Germany	1	1	0	1	0	3
	1.3529	0.5882	0.2941	0.4118	0.3529	
	0.98	0.98	0.00	0.98	0.00	2.94
	33.33	33.33	0.00	33.33	0.00	
	2.17	5.00	0.00	7.14	0.00	
Greece	0	0	0	1	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.00	0.00	0.98	0.00	0.98
	0.00	0.00	0.00	100.00	0.00	
	0.00	0.00	0.00	7.14	0.00	
India	3	0	0	0	0	3
	1.3529	0.5882	0.2941	0.4118	0.3529	
	2.94	0.00	0.00	0.00	0.00	2.94
	100.00	0.00	0.00	0.00	0.00	
	6.52	0.00	0.00	0.00	0.00	
Italy	35	9	7	9	5	65
	29.314	12.745	6.3725	8.9216	7.6471	
	34.31	8.82	6.86	8.82	4.90	63.73
	53.85	13.85	10.77	13.85	7.69	
	76.09	45.00	70.00	64.29	41.67	
Japan	0	1	0	1	0	2
	0.902	0.3922	0.1961	0.2745	0.2353	
	0.00	0.98	0.00	0.98	0.00	1.96
	0.00	50.00	0.00	50.00	0.00	
	0.00	5.00	0.00	7.14	0.00	
Lithuania	3	6	1	1	4	15
	6.7647	2.9412	1.4706	2.0588	1.7647	
	2.94	5.88	0.98	0.98	3.92	14.71
	20.00	40.00	6.67	6.67	26.67	
	6.52	30.00	10.00	7.14	33.33	
Netherlands	0	0	0	0	1	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.00	0.00	0.00	0.98	0.98
	0.00	0.00	0.00	0.00	100.00	
	0.00	0.00	0.00	0.00	8.33	
Poland	0	0	1	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.00	0.00	0.98	0.00	0.00	0.98
	0.00	0.00	100.00	0.00	0.00	
	0.00	0.00	10.00	0.00	0.00	
Romania	1	0	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.98	0.00	0.00	0.00	0.00	0.98
	100.00	0.00	0.00	0.00	0.00	
	2.17	0.00	0.00	0.00	0.00	
Russia	1	0	0	0	0	1
	0.451	0.1961	0.098	0.1373	0.1176	
	0.98	0.00	0.00	0.00	0.00	0.98
	100.00	0.00	0.00	0.00	0.00	
	2.17	0.00	0.00	0.00	0.00	
United Kingdom	0	1	0	0	2	3
	1.3529	0.5882	0.2941	0.4118	0.3529	
	0.00	0.98	0.00	0.00	1.96	2.94
	0.00	33.33	0.00	0.00	66.67	
	0.00	5.00	0.00	0.00	16.67	
United States	0	0	0	0	0	0
	0	0	0	0	0	
	0.00	0.00	0.00	0.00	0.00	0.00
	.	.	.	.	.	
	0.00	0.00	0.00	0.00	0.00	
Total	46	20	10	14	12	102
	45.10	19.61	9.80	13.73	11.76	100.00

- INCOME

Frequency
Expected
Percent
Row Pct
Col Pct

Table of d_7 by CLUSTER						
	CLUSTER					
d_7(d_7)	1	2	3	4	5	Total
1000 -2000	14 15.333 13.73 41.18 30.43	8 6.6667 7.84 23.53 40.00	3 3.3333 2.94 8.82 30.00	3 4.6667 2.94 8.82 21.43	6 4 5.88 17.65 50.00	34 33.33
2001-3000	9 8.1176 8.82 50.00 19.57	4 3.5294 3.92 22.22 20.00	2 1.7647 1.96 11.11 20.00	2 2.4706 1.96 11.11 14.29	1 2.1176 0.98 5.56 8.33	18 17.65
3001-4000	9 7.6667 8.82 52.94 19.57	3 3.3333 2.94 17.65 15.00	1 1.6667 0.98 5.88 10.00	2 2.3333 1.96 11.76 14.29	2 2 1.96 11.76 16.67	17 16.67
< 1000	11 8.5686 10.78 57.89 23.91	1 3.7255 0.98 5.26 5.00	3 1.8627 2.94 15.79 30.00	3 2.6078 2.94 15.79 21.43	1 2.2353 0.98 5.26 8.33	19 18.63
> 4001	3 6.3137 2.94 21.43 6.52	4 2.7451 3.92 28.57 20.00	1 1.3725 0.98 7.14 10.00	4 1.9216 3.92 28.57 28.57	2 1.6471 1.96 14.29 16.67	14 13.73
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- FAMILY MEMBERS

Frequency
Expected
Percent
Row Pct
Col Pct

Table of d_8 by CLUSTER						
	CLUSTER					
d_8(d_8)	1	2	3	4	5	Total
1	3 4.1461 3.37 33.33 7.32	2 1.618 2.25 22.22 12.50	2 0.7079 2.25 22.22 28.57	2 1.3146 2.25 22.22 15.38	0 1.2135 0.00 0.00 0.00	9 10.11
2	13 8.2921 14.61 72.22 31.71	0 3.236 0.00 0.00 0.00	0 1.4157 0.00 0.00 0.00	2 2.6292 2.25 11.11 15.38	3 2.427 3.37 16.67 25.00	18 20.22
3	6 9.2135 6.74 30.00 14.63	5 3.5955 5.62 25.00 31.25	3 1.573 3.37 15.00 42.86	1 2.9213 1.12 5.00 7.69	5 2.6966 5.62 25.00 41.67	20 22.47
4	19 19.348 21.35 45.24 46.34	9 7.5506 10.11 21.43 56.25	2 3.3034 2.25 4.76 28.57	8 6.1348 8.99 19.05 61.54	4 5.6629 4.49 9.52 33.33	42 47.19
Total	41 46.07	16 17.98	7 7.87	13 14.61	12 13.48	89 100.00

- MINORS IN THE FAMILY

Frequency Expected Percent Row Pct Col Pct	Table of d_9 by CLUSTER					
	CLUSTER					
d_9(d_9)	1	2	3	4	5	Total
No	39 33.824 38.24 52.00 84.78	9 14.706 8.82 12.00 45.00	8 7.3529 7.84 10.67 80.00	11 10.294 10.78 14.67 78.57	8 8.8235 7.84 10.67 66.67	75 73.53
Yes	7 12.176 6.86 25.93 15.22	11 5.2941 10.78 40.74 55.00	2 2.6471 1.96 7.41 20.00	3 3.7059 2.94 11.11 21.43	4 3.1765 3.92 14.81 33.33	27 26.47
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- STREAMING SERVICES USED

The FREQ Procedure

Frequency
Expected
Percent
Row Pct
Col Pct

Table of b_1 by CLUSTER	CLUSTER					
b_1(b_1)	1	2	3	4	5	Total
Amazon Prime Video, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Amazon Prime Video, NowTV, Illegal streaming platforms	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Anime website	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Hulu	0 0 0.00 0.00	0 0 0.00 0.00	0 0 0.00 0.00	0 0 0.00 0.00	0 0 0.00 0.00	0 0.00
Illegal streaming platforms	1 2.2549 0.98 20.00 2.17	0 0.9804 0.00 0.00 0.00	3 0.4902 2.94 60.00 30.00	1 0.6863 0.98 20.00 7.14	0 0.5882 0.00 0.00 0.00	5 4.90
Illegal streaming platforms, Dowload by Torrent	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Infinity	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix	2 7.6667 1.96 11.76 4.35	10 3.3333 9.80 58.82 50.00	0 1.6667 0.00 0.00 0.00	0 2.3333 0.00 0.00 0.00	5 2 4.90 29.41 41.67	17 16.67
Netflix, Amazon Prime Video	6 4.5098 5.88 60.00 13.04	1 1.9608 0.98 10.00 5.00	0 0.9804 0.00 0.00 0.00	1 1.3725 0.98 10.00 7.14	2 1.1765 1.96 20.00 16.67	10 9.80
Netflix, Amazon Prime Video, Disney +	8 8.5686 7.84 42.11 17.39	4 3.7255 3.92 21.05 20.00	1 1.8627 0.98 5.26 10.00	4 2.6078 3.92 21.05 28.57	2 2.2353 1.96 10.53 16.67	19 18.63
Netflix, Amazon Prime Video, Disney +, Crunchyroll	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, DAZN	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, DAZN, RaiPlay, sky go, Mediaset infinity	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Dazn, Infinity+	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, HBO Max	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, HBO Max, Hulu, NowTV, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, HBO Max, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Hulu	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Illegal streaming platforms	9 4.5098 8.82 90.00 19.57	0 1.9608 0.00 0.00 0.00	1 0.9804 0.98 10.00 10.00	0 1.3725 0.00 0.00 0.00	0 1.1765 0.00 0.00 0.00	10 9.80
Netflix, Amazon Prime Video, Disney +, Illegal streaming platforms, Twitch	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, NowTV	2 1.3529 1.96 66.67 4.35	0 0.5882 0.00 0.00 0.00	0 0.2941 0.00 0.00 0.00	0 0.4118 0.00 0.00 0.00	1 0.3529 0.98 33.33 8.33	3 2.94
Netflix, Amazon Prime Video, Disney +, NowTV, Illegal streaming platforms	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, Paramount plus	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Disney +, TimVision	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Hulu, U-NEXT	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, Illegal streaming platforms	2 1.3529 1.96 66.67 4.35	0 0.5882 0.00 0.00 0.00	0 0.2941 0.00 0.00 0.00	0 0.4118 0.00 0.00 0.00	1 0.3529 0.98 33.33 8.33	3 2.94
Netflix, Amazon Prime Video, Illegal streaming platforms, SkyGo	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, NowTV	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Amazon Prime Video, mubi	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Disney +	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Illegal streaming platforms	5 2.7059 4.90 83.33 10.87	0 1.1765 0.00 0.00 0.00	0 0.5882 0.00 0.00 0.00	0 0.8235 0.00 0.00 0.00	1 0.7059 0.98 16.67 8.33	6 5.88
Netflix, Illegal streaming platforms, * I used Netflix before, but now it doesn't work in my country, so I use just some streaming services of my country??	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Illegal streaming platforms, Mubi	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Prime video	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Netflix, Žmones cinema, various cinema festival sites (Ee. Kino pavasaris, NKdoku)	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
None	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Sky Go	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Telegram	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- TIME SPENT ON LEGAL PLATFORM

Sample Size = 102
Frequency Missing = 9

Frequency
Expected
Percent
Row Pct
Col Pct

Table of b_2 by CLUSTER						
	CLUSTER					
b_2(b_2)	1	2	3	4	5	Total
1-4 hours	13 18.49 12.75 31.71 28.26	5 8.0392 4.90 12.20 25.00	4 4.0196 3.92 9.76 40.00	11 5.6275 10.78 26.83 78.57	8 4.8235 7.84 19.51 66.67	41 40.20
30 mins	3 3.1569 2.94 42.86 6.52	2 1.3725 1.96 28.57 10.00	2 0.6863 1.96 28.57 20.00	0 0.9608 0.00 0.00 0.00	0 0.8235 0.00 0.00 0.00	7 6.86
5+ hours	27 20.745 26.47 58.70 58.70	12 9.0196 11.76 26.09 60.00	1 4.5098 0.98 2.17 10.00	2 6.3137 1.96 4.35 14.29	4 5.4118 3.92 8.70 33.33	46 45.10
I don't use streaming services	3 3.6078 2.94 37.50 6.52	1 1.5686 0.98 12.50 5.00	3 0.7843 2.94 37.50 30.00	1 1.098 0.98 12.50 7.14	0 0.9412 0.00 0.00 0.00	8 7.84
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
Frequency Missing = 9						

- TIME SPENT ON NETFLIX

Sample Size = 102
Frequency Missing = 9

Frequency
Expected
Percent
Row Pct
Col Pct

Table of b_3 by CLUSTER						
	CLUSTER					
b_3(b_3)	1	2	3	4	5	Total
1-4 hours	27 23.451 26.47 51.92 58.70	5 10.196 4.90 9.62 25.00	4 5.098 3.92 7.69 40.00	8 7.1373 7.84 15.38 57.14	8 6.1176 7.84 15.38 66.67	52 50.98
30 mins	1 3.1569 0.98 14.29 2.17	3 1.3725 2.94 42.86 15.00	0 0.6863 0.00 0.00 0.00	2 0.9608 1.96 28.57 14.29	1 0.8235 0.98 14.29 8.33	7 6.86
5+ hours	13 13.078 12.75 44.83 28.26	10 5.6863 9.80 34.48 50.00	0 2.8431 0.00 0.00 0.00	3 3.9804 2.94 10.34 21.43	3 3.4118 2.94 10.34 25.00	29 28.43
I don't watch Netflix	5 6.3137 4.90 35.71 10.87	2 2.7451 1.96 14.29 10.00	6 1.3725 5.88 42.86 60.00	1 1.9216 0.98 7.14 7.14	0 1.6471 0.00 0.00 0.00	14 13.73
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
Frequency Missing = 9						

- TIME SPENT ON ILLEGAL PLATFORMS

Sample Size = 102
Frequency Missing = 9

Frequency Expected Percent Row Pct Col Pct	Table of b_4 by CLUSTER					
		CLUSTER				
b_4(b_4)	1	2	3	4	5	Total
1-4 hours	14 10.373 13.73 60.87 30.43	0 4.5098 0.00 0.00 0.00	5 2.2549 4.90 21.74 50.00	0 3.1569 0.00 0.00 0.00	4 2.7059 3.92 17.39 33.33	23 22.55
30 mins	7 6.3137 6.86 50.00 15.22	2 2.7451 1.96 14.29 10.00	2 1.3725 1.96 14.29 20.00	1 1.9216 0.98 7.14 7.14	2 1.6471 1.96 14.29 16.67	14 13.73
5+ hours	5 4.0588 4.90 55.56 10.87	0 1.7647 0.00 0.00 0.00	2 0.8824 1.96 22.22 20.00	1 1.2353 0.98 11.11 7.14	1 1.0588 0.98 11.11 8.33	9 8.82
I don't use illegal streaming services	20 25.255 19.61 35.71 43.48	18 10.98 17.65 32.14 90.00	1 5.4902 0.98 1.79 10.00	12 7.6863 11.76 21.43 85.71	5 6.5882 4.90 8.93 41.67	56 54.90
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
Frequency Missing = 9						

- REASONS FOR USING ILLEGAL STREAMING

Sample Size = 102
Frequency Missing = 9

Frequency Expected Percent Row Pct Col Pct	Table of b_5 by CLUSTER					
		CLUSTER				
b_5(b_5)	1	2	3	4	5	Total
Better choices compared to Netflix, Disney+ or other legal streaming services	18 15.882 35.29 66.67 60.00	0 0.5294 0.00 0.00 0.00	3 4.7647 5.88 11.11 33.33	4 3.1765 7.84 14.81 66.67	2 2.6471 3.92 7.41 40.00	27 52.94
I randomly watch series and movies so I don't see the need to have a monthly subscription	4 4.7059 7.84 50.00 13.33	0 0.1569 0.00 0.00 0.00	3 1.4118 5.88 37.50 33.33	1 0.9412 1.96 12.50 16.67	0 0.7843 0.00 0.00 0.00	8 15.69
In my country legal streaming services are not available	3 1.7647 5.88 100.00 10.00	0 0.0588 0.00 0.00 0.00	0 0.5294 0.00 0.00 0.00	0 0.3529 0.00 0.00 0.00	0 0.2941 0.00 0.00 0.00	3 5.88
Prices for legal streaming services are too high	5 7.6471 9.80 38.46 16.67	1 0.2549 1.96 7.69 100.00	3 2.2941 5.88 23.08 33.33	1 1.5294 1.96 7.69 16.67	3 1.2745 5.88 23.08 60.00	13 25.49
Total	30 58.82	1 1.96	9 17.65	6 11.76	5 9.80	51 100.00
Frequency Missing = 60						

- SHARING OF THE ACCOUNT

Frequency Expected Percent Row Pct Col Pct	Table of b_6 by CLUSTER					
	CLUSTER					
b_6(b_6)	1	2	3	4	5	Total
Family	24 25.706 23.53 42.11 52.17	13 11.176 12.75 22.81 65.00	3 5.5882 2.94 5.26 30.00	7 7.8235 6.86 12.28 50.00	10 6.7059 9.80 17.54 83.33	57 55.88
Friends	17 12.627 16.67 60.71 36.96	3 5.4902 2.94 10.71 15.00	1 2.7451 0.98 3.57 10.00	6 3.8431 5.88 21.43 42.86	1 3.2941 0.98 3.57 8.33	28 27.45
I don't have an account	4 5.8627 3.92 30.77 8.70	2 2.549 1.96 15.38 10.00	6 1.2745 5.88 46.15 60.00	1 1.7843 0.98 7.69 7.14	0 1.5294 0.00 0.00 0.00	13 12.75
Nobody	1 1.8039 0.98 25.00 2.17	2 0.7843 1.96 50.00 10.00	0 0.3922 0.00 0.00 0.00	0 0.549 0.00 0.00 0.00	1 0.4706 0.98 25.00 8.33	4 3.92
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
	Frequency Missing = 9					

- WATCH PREFERENCE

Sample Size = 102
Frequency Missing = 9

Frequency Expected Percent Row Pct Col Pct	Table of b_7 by CLUSTER					
	CLUSTER					
b_7(b_7)	1	2	3	4	5	Total
Movies	8 12.627 7.84 28.57 17.39	7 5.4902 6.86 25.00 35.00	3 2.7451 2.94 10.71 30.00	7 3.8431 6.86 25.00 50.00	3 3.2941 2.94 10.71 25.00	28 27.45
Tv series	38 33.373 37.25 51.35 82.61	13 14.51 12.75 17.57 65.00	7 7.2549 6.86 9.46 70.00	7 10.157 6.86 9.46 50.00	9 8.7059 8.82 12.16 75.00	74 72.55
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
	Frequency Missing = 9					

- GENRES

Table of b_8 by CLUSTER

Frequency Expected Percent Row Pct Col Pct	b_8(b_8)	CLUSTER 1	2	3	4	5	Total
	action, comedy	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	action, comedy, crime, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
	action, comedy, dramas, crime, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
	action, comedy, dramas, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	action, comedy, fantasy, sci-fi, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
	action, comedy, romance	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	action, dramas, crime	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
	action, romance, thriller	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
	adventure, action, comedy, dramas, fantasy, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	adventure, action, comedy, war, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	adventure, action, dramas fantasy, crime, war, historical, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	adventure, action, dramas musicals experimental, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	adventure, action, fantasy, war, historical, sci-fi, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	adventure, comedy, crime, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	adventure, comedy, dramas	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
	adventure, comedy, dramas fantasy, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
	adventure, comedy, dramas romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98

adventure, comedy, fantasy, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
adventure, crime, sci-fi	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
adventure, fantasy, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, action, comedy, fantasy, musicals, sci-fi, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, action, dramas, fantasy, crime, war, historical, romance	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
animation, adventure, action, comedy, dramas, fantasy, crime, sci-fi, thriller	1 0.451 0.98	0 0.1961 0.00	0 0.098 0.00	0 0.1373 0.00	0 0.1176 0.00	1 0.98

	. 0.00	. 0.00	. 0.00	. 0.00	. 0.00	
animation, adventure, action, comedy, dramas, fantasy, crime, sci-fi, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, action, comedy, dramas, sci-fi, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, action, comedy, fantasy, historical	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, action, comedy, fantasy, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, action, comedy, musicals, sci-fi, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, action, comedy, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, action, dramas, crime, musicals, romance	0	1	0	0	0	1

	0.00 0.00 0.00	0.00 0.00 0.00	0.98 100.00 10.00	0.00 0.00 0.00	0.00 0.00 0.00	0.98
animation, adventure, action, dramas, crime, musicals, romance	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, comedy, dramas, musicals	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, adventure, comedy, fantasy, historical	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
animation, comedy, dramas, fantasy, romance	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
animation, comedy, fantasy, historical, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
comedy	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98

	0.451 0.00 0.00 0.00	0.1961 0.00 0.00 0.00	0.098 0.98 100.00 10.00	0.1373 0.00 0.00 0.00	0.1176 0.00 0.00 0.00	0.98
comedy, crime	1 0.902 0.98 50.00 2.17	0 0.3922 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.2745 0.98 50.00 7.14	0 0.2353 0.00 0.00 0.00	2 1.96
comedy, dramas	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
comedy, dramas, crime	1 0.902 0.98 50.00 2.17	1 0.3922 0.98 50.00 5.00	0 0.1961 0.00 0.00 0.00	0 0.2745 0.00 0.00 0.00	0 0.2353 0.00 0.00 0.00	2 1.96
comedy, dramas, crime, war, sports, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
comedy, dramas, historical, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
comedy, dramas, musicals	0 0.451 0.00 0.00	1 0.1961 0.98 100.00	0 0.098 0.00 0.00	0 0.1373 0.00 0.00	0 0.1176 0.00 0.00	1 0.98

comedy, dramas, historical, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
comedy, dramas, musicals	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
comedy, fantasy, crime, musicals, historical, sci-fi, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
comedy, romance	1 0.902 0.98 50.00 2.17	0 0.3922 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.2745 0.98 50.00 7.14	0 0.2353 0.00 0.00 0.00	2 1.96
comedy, sci-fi, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
comedy, sports, thriller	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
crime, war, historical, sci-fi, romance, thriller	0 0.451 0.00	0 0.1961 0.00	0 0.098 0.00	1 0.1373 0.98	0 0.1176 0.00	1 0.98

	0.00	5.00	0.00	0.00	0.00	
crime, war, historical, sci-fi, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, action, comedy, dramas, crime, romance, thriller	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, action, crime, sci-fi, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, action, crime, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, adventure, action, comedy, crime	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, adventure, action, comedy, dramas, fantasy, sci-fi, romance, thriller	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, adventure, action, crime, sports	0 0.451	0 0.1961	0 0.098	0 0.1373	1 0.1176	1

	0.00 0.00	100.00 5.00	0.00 0.00	0.00 0.00	0.00 0.00	
documentaries, adventure, action, crime, sports	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
documentaries, adventure, action, historical, sports, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, adventure, comedy, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, adventure, dramas	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, adventure, dramas, fantasy, historical, experimental, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, adventure, action, comedy, dramas, fantasy, crime, war, historical, sci-fi, experimental, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, adventure, action, comedy, dramas, fantasy, crime, war, historical, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, adventure, action, comedy, dramas, fantasy, historical, sci-fi, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, adventure, action, comedy, dramas, war, musicals, historical, sci-fi, romance	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, adventure, comedy, dramas, fantasy, sci-fi, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, adventure, comedy, dramas, romance	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
documentaries, animation, adventure, comedy, fantasy, historical, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, comedy, dramas, fantasy	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, animation, dramas, experimental	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, comedy	0 0.902 0.00 0.00 0.00	1 0.3922 0.98 50.00 5.00	0 0.1961 0.00 0.00 0.00	0 0.2745 0.00 0.00 0.00	1 0.2353 0.98 50.00 8.33	2 1.96
documentaries, comedy, crime, historical	0 0.902 0.00 0.00 0.00	2 0.3922 1.96 100.00 10.00	0 0.1961 0.00 0.00 0.00	0 0.2745 0.00 0.00 0.00	0 0.2353 0.00 0.00 0.00	2 1.96
documentaries, comedy, crime, sci-fi, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, comedy, historical, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, crime	0 0	0 0	0 0	0 0	0 0	0

	0.00 . 0.00	0.00 . 0.00	0.00 . 0.00	0.00 . 0.00	0.00 . 0.00	0.00
documentaries, crime, historical, romance	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, action, comedy, crime, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
documentaries, horror, action, comedy, crime, war, romance	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, action, comedy, fantasy, crime, sci-fi, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, animation, action, comedy, fantasy, crime, thriller	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
documentaries, horror, animation, adventure, action, comedy, dramas, crime, romance, sports, thriller	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
	0.00	0.00	0.00	0.00	0.00	
documentaries, horror, animation, adventure, action, comedy, dramas, fantasy, crime, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, animation, adventure, action, comedy, fantasy, historical, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, animation, adventure, action, comedy, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, animation, adventure, comedy, fantasy	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, animation, adventure, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, animation, comedy, fantasy, musicals, historical	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, horror, crime, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	1 0.1176 0.98 100.00 8.33	1 0.98
documentaries, horror, dramas, historical, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
documentaries, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
dramas	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
dramas, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
fantasy	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
fantasy, crime, romance	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, adventure, action, fantasy, crime, war, experimental, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, adventure	0	0	0	0	1	1

	0.451 0.00 0.00 0.00	0.1961 0.00 0.00 0.00	0.098 0.00 0.00 0.00	0.1373 0.00 0.00 0.00	0.1176 0.98 100.00 8.33	0.98
horror, animation, adventure, action, comedy, dramas, fantasy, crime, musicals, sci-fi, romance, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, adventure, action, comedy, dramas, fantasy, crime, war, musicals, historical, sci-fi, experimental, romance, thriller	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	1 0.1373 0.98 100.00 7.14	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, adventure, action, comedy, dramas, fantasy, crime, war, sci-fi, sports, thriller	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, adventure, action, comedy, dramas, fantasy, sci-fi, romance	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, adventure, action, comedy, fantasy, crime, war, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, adventure, action, comedy, fantasy, sci-fi, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, adventure, comedy, fantasy, sci-fi	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, comedy	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, animation, historical, sci-fi, experimental, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, crime	0 0.451 0.00 0.00 0.00	0 0.1961 0.00 0.00 0.00	1 0.098 0.98 100.00 10.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, crime, sci-fi, experimental, thriller	1 0.451 0.98 100.00 2.17	0 0.1961 0.00 0.00 0.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
horror, dramas	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0 0.00 . 0.00	0 0.00
sci-fi	0 0.451 0.00 0.00 0.00	1 0.1961 0.98 100.00 5.00	0 0.098 0.00 0.00 0.00	0 0.1373 0.00 0.00 0.00	0 0.1176 0.00 0.00 0.00	1 0.98
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

- PRICE

The FREQ Procedure

Frequency Expected Percent Row Pct Col Pct

Table of I_11 by CLUSTER						
	CLUSTER					
I_11(I_11)	1	2	3	4	5	Total
0	11	6	2	1	1	21
	9.4706	4.1176	2.0588	2.8824	2.4706	
	10.78	5.88	1.96	0.98	0.98	20.59
	52.38	28.57	9.52	4.76	4.76	
	23.91	30.00	20.00	7.14	8.33	
1	35	14	8	13	11	81
	36.529	15.882	7.9412	11.118	9.5294	
	34.31	13.73	7.84	12.75	10.78	79.41
	43.21	17.28	9.88	16.05	13.58	
	76.09	70.00	80.00	92.86	91.67	
Total	46	20	10	14	12	102
	45.10	19.61	9.80	13.73	11.76	100.00
Frequency Missing = 9						

Statistics for Table of I_11 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	4.0470	0.3997
Likelihood Ratio Chi-Square	4	4.5852	0.3326
Mantel-Haenszel Chi-Square	1	2.7199	0.0991
Phi Coefficient		0.1992	
Contingency Coefficient		0.1954	
Cramer's V		0.1992	
WARNING: 40% of the cells have expected counts less than 5. Chi-Square may not be a valid test.			

- INDOOR vs OUTDOOR

Frequency Expected Percent Row Pct Col Pct

Table of I_22 by CLUSTER						
	CLUSTER					
I_22(I_22)	1	2	3	4	5	Total
0	11	4	4	3	0	22
	9.9216	4.3137	2.1569	3.0196	2.5882	
	10.78	3.92	3.92	2.94	0.00	21.57
	50.00	18.18	18.18	13.64	0.00	
	23.91	20.00	40.00	21.43	0.00	
1	35	16	6	11	12	80
	36.078	15.686	7.8431	10.98	9.4118	
	34.31	15.69	5.88	10.78	11.76	78.43
	43.75	20.00	7.50	13.75	15.00	
	76.09	80.00	60.00	78.57	100.00	
Total	46	20	10	14	12	102
	45.10	19.61	9.80	13.73	11.76	100.00
Frequency Missing = 9						

Statistics for Table of I_22 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	5.4869	0.2409
Likelihood Ratio Chi-Square	4	7.7328	0.1019
Mantel-Haenszel Chi-Square	1	1.3661	0.2425
Phi Coefficient		0.2319	
Contingency Coefficient		0.2259	
Cramer's V		0.2319	
WARNING: 40% of the cells have expected counts less than 5. Chi-Square may not be a valid test.			

- HEALTHY LIFESTYLE

Frequency Expected Percent Row Pct Col Pct	Table of I_33 by CLUSTER					
	CLUSTER					
I_33(I_33)	1	2	3	4	5	Total
0	12 14.431 11.76 37.50 26.09	9 6.2745 8.82 28.13 45.00	4 3.1373 3.92 12.50 40.00	3 4.3922 2.94 9.38 21.43	4 3.7647 3.92 12.50 33.33	32 31.37
1	34 31.569 33.33 48.57 73.91	11 13.725 10.78 15.71 55.00	6 6.8627 5.88 8.57 60.00	11 9.6078 10.78 15.71 78.57	8 8.2353 7.84 11.43 66.67	70 68.63
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
	Frequency Missing = 9					

Statistics for Table of I_33 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	3.3321	0.5039
Likelihood Ratio Chi-Square	4	3.2829	0.5116
Mantel-Haenszel Chi-Square	1	0.0320	0.8580
Phi Coefficient		0.1807	
Contingency Coefficient		0.1779	
Cramer's V		0.1807	
WARNING: 30% of the cells have expected counts less than 5. Chi-Square may not be a valid test.			

- IMPORTANCE OF LEISURE TIME

Frequency Expected Percent Row Pct Col Pct	Table of I_44 by CLUSTER					
	CLUSTER					
I_44(I_44)	1	2	3	4	5	Total
0	7 6.3137 6.86 50.00 15.22	4 2.7451 3.92 28.57 20.00	1 1.3725 0.98 7.14 10.00	1 1.9216 0.98 7.14 7.14	1 1.6471 0.98 7.14 8.33	14 13.73
1	39 39.686 38.24 44.32 84.78	16 17.255 15.69 18.18 80.00	9 8.6275 8.82 10.23 90.00	13 12.078 12.75 14.77 92.86	11 10.353 10.78 12.50 91.67	88 86.27
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00
	Frequency Missing = 9					

Statistics for Table of I_44 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	1.6755	0.7952
Likelihood Ratio Chi-Square	4	1.7484	0.7819
Mantel-Haenszel Chi-Square	1	0.9238	0.3365
Phi Coefficient		0.1282	
Contingency Coefficient		0.1271	
Cramer's V		0.1282	
WARNING: 40% of the cells have expected counts less than 5. Chi-Square may not be a valid test.			

- RISK-TAKER

Frequency
Expected
Percent
Row Pct
Col Pct

Table of I_55 by CLUSTER

	CLUSTER					
I_55(I_55)	1	2	3	4	5	Total
0	29 25.706 28.43 50.88 63.04	8 11.176 7.84 14.04 40.00	7 5.5882 6.86 12.28 70.00	9 7.8235 8.82 15.79 64.29	4 6.7059 3.92 7.02 33.33	57 55.88
1	17 20.294 16.67 37.78 36.96	12 8.8235 11.76 26.67 60.00	3 4.4118 2.94 6.67 30.00	5 6.1765 4.90 11.11 35.71	8 5.2941 7.84 17.78 66.67	45 44.12
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

Frequency Missing = 9

Statistics for Table of I_55 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	6.6874	0.1534
Likelihood Ratio Chi-Square	4	6.7211	0.1514
Mantel-Haenszel Chi-Square	1	1.1062	0.2929
Phi Coefficient		0.2561	
Contingency Coefficient		0.2481	
Cramer's V		0.2561	

- STRESSED

Frequency
Expected
Percent
Row Pct
Col Pct

Table of I_66 by CLUSTER

	CLUSTER					
I_66(I_66)	1	2	3	4	5	Total
0	17 15.784 16.67 48.57 36.96	8 6.8627 7.84 22.86 40.00	2 3.4314 1.96 5.71 20.00	6 4.8039 5.88 17.14 42.86	2 4.1176 1.96 5.71 16.67	35 34.31
1	29 30.216 28.43 43.28 63.04	12 13.137 11.76 17.91 60.00	8 6.5686 7.84 11.94 80.00	8 9.1961 7.84 11.94 57.14	10 7.8824 9.80 14.93 83.33	67 65.69
Total	46 45.10	20 19.61	10 9.80	14 13.73	12 11.76	102 100.00

Frequency Missing = 9

Statistics for Table of I_66 by CLUSTER

Statistic	DF	Value	Prob
Chi-Square	4	3.4498	0.4856
Likelihood Ratio Chi-Square	4	3.7253	0.4445
Mantel-Haenszel Chi-Square	1	0.9035	0.3418
Phi Coefficient		0.1839	
Contingency Coefficient		0.1809	
Cramer's V		0.1839	

WARNING: 30% of the cells have expected counts less than 5. Chi-Square may not be a valid test.